

# CORPORATE DIGITAL TRANSFORMATION AND ENVIRONMENTAL PERFORMANCE

EVIDENCE FROM THE PEOPLE'S REPUBLIC OF CHINA

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# Corporate Digital Transformation and Environmental Performance: Evidence from the People's Republic of China

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## ABSTRACT

This paper examines the impact of corporate digital transformation on environmental performance and the underlying mechanisms. Utilizing textual analysis of annual reports, we develop a four-dimensional measure of digital transformation comprising technological, business, manufacturing, and operational aspects. Analyzing data from Chinese A-share listed firms between 2009 and 2022, we demonstrate that corporate digital transformation significantly enhances environmental performance. While all dimensions of digital transformation contribute positively, manufacturing digital transformation exhibits the most substantial impact, followed by operational and technological digital transformation. We identify that green innovation, strengthened environmental monitoring and communication, increased environmental information transparency, and greater external attention are pivotal mechanisms through which digital transformation affects environmental performance, with transparency emerging as the strongest factor in this context. This paper offers new insights into the role of digital transformation in advancing sustainability.

**Keywords:** corporate digital transformation, environmental performance, textual analysis, innovation, transparency

**JEL codes:** D22, O33, Q56

## I. INTRODUCTION

In the digital era, the digital revolution emerged as a critical driver for achieving the Sustainable Development Goals (United Nations 2020). Countries are increasingly formulating policies that promote the interplay between digitalization and sustainability, both of which are among the most powerful market influences in today's corporate landscape. However, diverse practices lead to varying outcomes, and the effectiveness of this synergy remains a subject of active debate and research.

A growing body of academic literature has explored the impact of digitalization on sustainability. Numerous studies indicate that the digital economy enhances resource efficiency (Dubey et al. 2019, Ren et al. 2021, Wang et al. 2022), improves green total factor productivity (Lee et al. 2023), reduces carbon emissions (Wang et al. 2022, Kopp et al. 2023), drives high-quality green development (ElMassah and Mohieldin 2020, Ma and Zhu 2022), increases resilience (Copestake et al. 2024), and fosters green economic growth (Hao et al. 2023). Recently, a new stream of research has begun to investigate the real effects of digitalization on corporate sustainable development, such as carbon emissions reduction (Zhou and Liu 2024, Liu et al. 2024), environmental performance improvement (Xu et al. 2023, Song et al. 2024), and environmental, social, and governance (ESG) ratings enhancement (Zhao and Cai 2023, Li et al. 2024). Our study will enrich this line of literature by examining whether and how corporate digital transformation influences environmental performance.

Theoretically, the relationship between corporate digital transformation and environmental performance is ambiguous. On the one hand, the application of digital technologies enables firms to reduce the search, replication, transportation, tracking, and verification costs of data (Borgatti and Cross 2003, Goldfarb and Tucker 2019); overcome knowledge localization barriers (Jaffe et al. 1993, Forman and Van Zeebroeck 2019); enhance communication (Nguyen et al. 2020); support the development of new offerings, processes, or solutions (Tortora et al. 2021); strengthen the collection and monitoring of environmental data (Chapron 2017); and thereby improve management capabilities across all aspects of corporate operations (Graetz and Michaels 2018, Fromhold-Eisebith et al. 2021, Chen and Jiang 2024).

On the other hand, it is important to acknowledge the potential downsides of digitalization. First, corporate digital transformation entails significant upfront investment and maintenance costs, and it can increase energy consumption and e-waste generation (Ren et al. 2021). Technology misalignment may lead to ineffective or even failed digital investments. Additionally, political, social, and psychological concerns surrounding digitalization are increasingly highlighted (Trittin-Ulbrich et al. 2021), such as inequality, employment displacement, psychological dependence, and technological reliance. Nevertheless, there remains widespread optimism about digital transformation, with ongoing efforts to explore its potential to enhance environmental performance.

Identifying corporate digital transformation levels accurately is challenging because of the absence of a clearly defined or universally accepted evaluation standard. Verhoef et al. (2021) propose that firms can be categorized into three stages: (i) digitization, (ii) digitalization, and (iii) digital transformation, based on their organizational structure, digital assets, and capabilities. They define digital transformation as the process through which enterprises leverage advanced digital technologies, such as big data, artificial intelligence (AI), and industrial robotics, to transform production processes, business models, internal controls, and organizational culture, thereby adapting to technological advancements and achieving competitive advantage. Inspired by Verhoef et al. (2021), we focus on "the application of digital technologies in enterprises" and aim to construct a more comprehensive and structured measure of digital transformation. Drawing from classical functional domain theories in management, we develop a four-dimensional digital transformation measure, including technological, business, manufacturing, and operational

aspects. Using textual analysis, we analyze the annual reports of listed companies to extract keyword frequencies corresponding to each dimension, thereby obtaining both individual and overall digital transformation levels. This structured approach provides a clear foundation for subsequent empirical analysis based on the defined dimensions.

Our primary analysis investigates the impact of corporate digital transformation on environmental performance using a large sample of Chinese A-share listed firms from 2009 to 2022. This study employs the environmental performance score (*EScore*) developed by the Sino-Securities Index (SNSI) ESG ratings as the dependent variable, with the constructed digital transformation levels serving as the core independent variable. Our baseline results indicate that an increase in digital transformation exerts a positive and statistically significant effect on environmental performance, after controlling for firm-specific characteristics and incorporating firm and year fixed effects. Specifically, all else being equal, a one standard deviation increase in corporate digital transformation corresponds to a 1.34% enhancement in environmental performance in this context.

To address the issue of endogeneity, we first utilize instrumental variable (IV) estimations constructed from the digital transformation levels of peer firms within the same industry, employing a two-stage least squares (TSLS) method. Additionally, we conduct a difference-in-differences (DID) analysis, using the “Broadband China” strategy as an exogenous shock (Tang et al. 2021). Launched in 2013, this initiative by the Government of the People’s Republic of China (PRC) aims to develop a nationwide high-speed broadband network and foster widespread adoption of information technology. The strategy was implemented in phases across 120 pilot cities in 2014, 2015, and 2016. The evidence suggests that city-level digitalization improves corporate environmental performance, helping establish a causal relationship between digital transformation and environmental performance.

The results remain robust when employing alternative environmental performance measures, including the *EScore* from the Wind ESG ratings and corporate environmental protection tax. Further validation using a change-on-change model specification, as well as concurrent and lagged digital transformation measures, confirm the consistency of our findings. Moreover, excluding financial, information services, and special treatment-listed firms does not affect our results.

Corporate digital transformation can influence environmental performance through several plausible channels. First of all, the most well-documented channel is green innovation. Numerous studies indicate that digitalization facilitates green innovation (e.g., Horbach 2008, Tang et al. 2023, Du et al. 2023), which in turn enhances environmental performance (Singh et al. 2020). On the one hand, digital technologies improve individual research and development (R&D) capabilities and promote knowledge spillover (Trantopoulos et al. 2017, Zhuo and Chen 2023). For instance, AI and big data enhance analytical capabilities (Dubey et al. 2019), while industrial internet and blockchain provide accurate and timely data (Chapron 2017). On the other hand, digital transformation significantly alters communication and cooperation models within and outside organizations, promoting R&D collaboration across the value chain (Benzidia et al. 2021). Wu et al. (2019) show that data analytics facilitate decentralized innovation. De Marchi (2012) highlights that R&D cooperation with suppliers, knowledge-intensive business services, and universities is essential for firms aiming to achieve environmental innovation. We term this channel as the *green innovation* channel.

The second plausible channel through which digital transformation enhances environmental performance is suggested by few theoretical models on environmental management (e.g., Klassen and McLaughlin 1996, Melnyk et al. 2003). In these models, environmental management—efforts to minimize the negative environmental impacts of a firm’s products

throughout their life cycle—has the potential to significantly influence environmental performance. The operations strategy literature indicates that both structural (plant and equipment) and infrastructural (production planning, performance management, and product design) choices require effective environmental monitoring and communication to support sound decision-making (Vogler and Eisenegger 2021).

Importantly, information is a prerequisite for assessing environmental conditions and making informed decisions to address identified challenges (Gholami et al. 2016). As extreme weather events increase, firms increasingly require timely responses to climate risks and uncertainties in environmental policy, necessitating adjustments in management strategies, such as investment and financing decisions (Benincasa et al. 2024). Additionally, changes in work practices brought about by digitalization, like virtual and hybrid conferences (Tao et al. 2021), facilitate more effective climate change mitigation efforts. We refer to this channel as the *environmental monitoring and communication* channel.

The third plausible channel is related to the information asymmetry problem and discretionary disclosure documented in the literature on transparency, with a specific focus on environmental information transparency. Numerous studies have confirmed that environmental information disclosure significantly affects firms' environmental performance (e.g., Patten 2002, Clarkson et al. 2008, Shi et al. 2021), primarily by strengthening external governance. Wang et al. (2023) demonstrate that information transmission channels affect firms' disclosure behaviors. As societal awareness and preferences for environmental sustainability grow, there is an increasing demand for greater transparency in environmental information across various industries. Hence, we term this channel as the *environmental information transparency* channel.

The fourth plausible channel at play involves external attention, complementing the third mechanism, as both promote the implementation of external oversight. The intensity of this oversight is closely linked to the level of attention firms receive, and external attention has been shown to be a powerful force driving firms to go green. Irani and Oesch (2013) demonstrate that analyst monitoring complements corporate governance and impacts financial reporting quality. Buntaine et al. (2024) show that public appeals through social media significantly reduce firms' pollution emissions.

A firm's ability to attract attention from local governments, analysts, media, investors, and the public is linked to its visibility and accessibility. Further, the effectiveness of external monitoring is closely related to the availability and quality of information. Liu et al. (2024) argue that governments and investors play a critical oversight role through which corporate digital transformation reduces carbon emissions. In addition to publicly disclosed information, nonpublic information that stakeholders can obtain through personal relationships is also crucial and can be enhanced by corporate digital transformation (Altschuler et al. 2015). We refer to this channel as the *external attention* channel.

Additionally, we evaluate the relative importance of these mechanisms using the horse race approach. We find that environmental information transparency has the strongest impact in this context, followed by green innovation and external attention. Although environmental monitoring and communication lacks significance in this analysis, this may be primarily attributed to our choice of measuring it through the establishment of environmental emergency response mechanisms; exploring a more comprehensive proxy could yield further insights.

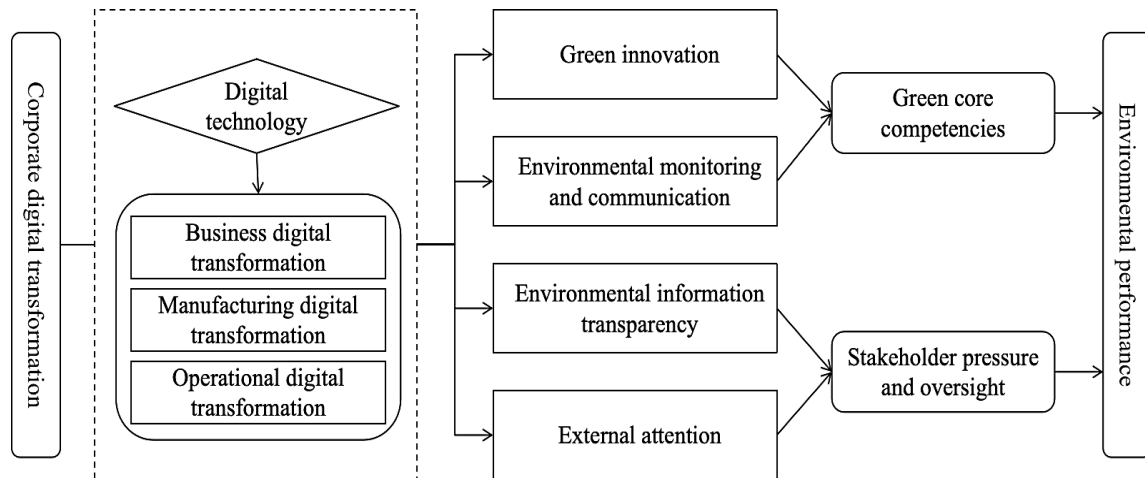
We further analyze whether the sub-dimensions of digital transformation exert differing impacts on environmental performance and assess their relative importance. The findings show that all digital transformation dimensions significantly improve environmental performance. Notably, under the horse race approach, manufacturing digital transformation demonstrates the strongest impact, followed by operational digital transformation and technological digital transformation,

while business digital transformation does not show significance. Our results indicate that, although all forms of digital transformation contribute to promoting sustainability, the benefits derived from specific leverage points vary. Merely altering business models yields limited advantages; effectively applying digital transformation to production and operational management, along with the appropriate utilization of digital technologies, is essential.

Lastly, we conduct a heterogeneity analysis on firm characteristics to investigate whether the relationship between digital transformation and environmental performance is driven by specific groups, thereby addressing concerns related to the digital divide. Our findings indicate that the positive impact of digital transformation on environmental performance holds across various firm sizes, ownership types, and regional economic development levels. However, the economic impact varies across the four sub-dimensions of digital transformation for firms with different characteristics. Therefore, both managers and governments should create tailored digital strategies or policies.

Overall, all of these findings suggest that there exists a positive, causal effect of corporate digital transformation on environmental performance. The key mechanisms include green innovation, environmental monitoring and communication, environmental information transparency, and external attention, as shown in Figure 1.

**Figure 1: Mechanism Analysis of Corporate Digital Transformation and Environmental Performance**



Source: Authors' illustrations.

Our paper makes several contributions to the growing literature on the impact of digitalization on sustainability. Previous studies have primarily focused on how the digital economy and technologies influence sustainable development at the macro level, with limited research on firm-level effects. Song et al. (2024) find that corporate digital transformation enhances environmental performance through green technological innovation, external media attention, and internal management control. Our paper contributes to this line of inquiry by examining the impact of corporate digital transformation on environmental performance and the underlying mechanisms.

First, we develop a four-dimensional measure of corporate digital transformation and analyze its role across technological, business, manufacturing, and operational aspects, assessing their respective impacts on firms' environmental performance. Our findings indicate that a one standard deviation increase in digital transformation corresponds to a 1.34% improvement in environmental performance. While all dimensions of digital transformation positively influence environmental performance, manufacturing digital transformation exhibits the strongest effect, followed by



operational and technological digital transformation. Further, the benefits derived from these four dimensions vary among different types of firms.

Second, we identify two new mechanisms—environmental monitoring and communication, and environmental information transparency—beyond the established mechanisms of green innovation and external attention. Additionally, we evaluate the relative importance of these four mechanisms, surprisingly revealing that transparency exerts the most pronounced influence in this context.

Finally, we enrich firm-level evidence by utilizing a sample with broader sector coverage. While several studies offer empirical evidence from specific representative industries (Xu et al. 2023, Zhou and Liu 2024), we demonstrate that the positive effect of digital transformation on environmental performance is consistent across various sectors. Our findings affirm the feasibility and effectiveness of leveraging digital transformation to promote environmental performance for corporations of all types, contingent upon the development of tailored strategies that take into account specific firm characteristics and contextual environments.

Our work contributes more broadly to the literature on corporate sustainable development (e.g., Russo and Fouts 1997, Hart 1997, Bansal and Roth 2000, Horbach 2008). Hart (1997) argues that technological change is essential for reducing environmental burdens. Bansal and Roth (2000) identify that legislation, stakeholder pressures, competitiveness, and leadership ecological responsibility are the key antecedents of corporate ecological responsiveness. Our findings contribute to this literature by reinforcing the critical roles of technological innovation and environmental pressures from stakeholders, while also highlighting the importance of environmental monitoring and communication, as well as information transparency for corporate sustainability.

The remainder of the paper is organized as follows. Section II reviews the relevant literature and formulates the hypotheses. Section III describes the data and methodologies. Section IV presents the empirical results, while Section V concludes.

## **II. LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT**

### **A. Linkage of Corporate Digital Transformation to Environmental Performance**

#### **1. Corporate Digital Transformation**

The concept of digital transformation has gained significant traction in contemporary business discourse, representing the disruptive effects of digital technologies—encompassing information, computing, communication, and connectivity—on various aspects of business operations. This transformation encompasses the emergence of new business strategies and models; innovative processes, products, and services; enhanced organizational capabilities; novel interfirm relationships within extended business networks; and evolving market behaviors (Verhoef et al. 2021, Tortora et al. 2021). More broadly, it signifies the necessity for established firms to undergo radical changes in openness, affordances, and generativity to thrive in the digital landscape (Nambisan et al. 2019).

The most direct and fundamental impact of digital transformation is evident in the literature on information and knowledge. Goldfarb and Tucker (2019) emphasize that digital technologies reduce the costs associated with data storage, computation, and transmission, leading to a decrease in five distinct economic costs: search, replication, transportation, tracking, and verification costs. The changes brought about by digital technologies encompass data sources, expanding from numbers and text to include images and voice (Gillies et al. 2016), while also

enhancing the efficiency of information processing, particularly the capacity to analyze vast amounts of unstructured data, and altering the ways we learn and utilize knowledge. This knowledge spillover associated with digital economic activities transcends geographic and technological barriers (Jaffe et al. 1993, Forman and Van Zeebroeck 2019), fostering new paradigms in innovation, management, organization, collaboration, competition, corporate governance, and public participation. In the realm of environmental applications, digital technologies such as blockchain and the Internet of Things (IoT) can provide traceable life-cycle “ecological footprints” for products and processes, thereby benefiting governance and sustainability (Chapron 2017).

The literature on digitalization explores the broad effects of digital technologies and digital economy. Vial (2021) reviews studies that highlight digital technologies as sources of disruption, including their influence on consumer behavior and expectations, the competitive landscape, and the necessity of leveraging these technologies to identify new avenues for value creation. Earlier research has examined the role of digitalization in enhancing innovation management (Nambisan et al. 2019, Teece 2018), promoting business models and performance (Di Vaio et al. 2021), reshaping employment and labor structures (Balsmeier and Woerter 2019, Cirillo et al. 2021), and improving economic resilience (Copestake et al. 2024). Overall, digital transformation is believed to contribute significantly to the long-term value creation process and to support global, inclusive growth.

## **2. Corporate Environmental Performance**

The management literature underscores sustainability, including eco-responsibility, as a crucial corporate obligation that offers both internal and external long-term benefits (e.g., McGuire et al. 1988, Eccles et al. 2014). The ongoing debate regarding the relationship between economic performance and corporate social responsibility (CSR), which has persisted since the 1960s, remains unresolved, encompassing positive perspectives (e.g., Waddock and Graves 1997), neutral views (e.g., McWilliams and Siegel 2001), and negative assessments (e.g., Baird et al. 2012).

Drawing on the resource-based view, many scholars have examined the determinants of corporate environmental performance. Bansal and Roth (2000) identify motivations and contextual factors that drive corporate ecological responsiveness, categorizing them as competitiveness, legitimation, and ecological responsibility. Horbach (2008) analyzes German data to explore the determinants of environmental innovations, highlighting the significance of knowledge capital, environmental management tools, and organizational changes in fostering such innovations. Overall, key determinants can be classified as internal factors—referred to as “green core competencies”—and external factors, termed “green pressure.” Hart (1997) introduces a sustainability portfolio, comprising sustainability vision, pollution prevention, product stewardship, and clean technology, to guide firms toward greener practices.

## **3. Nexus Between Digital Technologies and Environmental Performance**

The emergence of a diverse array of novel and powerful digital technologies, business applications, and infrastructures has significantly transformed both green core capabilities and the transmission of green pressures, creating substantial potential for improved environmental performance.

Earlier research on the relationship between digital and green transitions has primarily emphasized macro-level analyses or specific digital technologies. Studies have demonstrated

notable positive effects of the digital economy, such as reductions in energy consumption intensity (Ren et al. 2021, Wang et al. 2022), decreases in carbon emissions of firms and households (Wang et al. 2022, Kopp et al. 2023), and enhancements in green total factor productivity (Lee et al. 2023). High-quality green development with spatial spillover effects is predominantly achieved through industrial structure adjustments and green technology innovations (Ma and Zhu 2022, Hao et al. 2023). Notable digital technologies, including big data (Dubey et al. 2019), AI (Dahlke et al. 2024), blockchain (Li et al. 2022), 3D printing (Beltagui et al. 2020), and digital platforms (Carrasco-Farre et al. 2022), exhibit significant economic and environmental impacts.

Conversely, some scholars contend that the relationship can be uncertain. Given the initial costs required to achieve scale effects, Kopp et al. (2023) argue that green technology innovations may present a threshold effect, negatively impacting low-income countries. Ye et al. (2023) assert that the positive influence of digital investment on environmental performance is significant only in state-owned and heavy industrial firms, but not in private or light industrial firms.

At the firm level, several studies show the substantial contributions of corporate digital transformation to climate change mitigation (Song et al. 2024, Liu et al. 2024, Zhou and Liu 2024). Song et al. (2024) note that digital transformation positively impacts reductions in corporate environmental taxes, primarily through green technology upgrades, increased external media attention, and enhanced internal management controls. Liu et al. (2024) indicate that digital transformation helps firms enhance their carbon emissions reduction performance, highlighting the significance of strengthening supply chain connections, improving capacity utilization, and the moderating role of external oversight groups. Overall, research at the firm level remains limited, with much evidence confined to specific representative sectors or heavy polluters. To enrich firm-level evidence on this line of literature, this paper examines the proposition that digital transformation influences a firm's environmental performance. We propose the hypothesis as follows:

**Hypothesis 1.** *Ceteris paribus, corporate digital transformation exerts a significant positive impact on environmental performance.*

## **B. Possible Mechanisms**

The stakeholder theory and sustainable development theory suggest that enterprises aiming for long-term growth must consider the environmental impacts of their manufacturing processes and products, as well as initiatives related to environmental management and technology, compliance with environmental regulations, and society's demands for environmental protection. Corporate digital transformation, therefore, is a significant component of functional strategies that enhance green core capabilities and foster environmental consciousness. Consequently, the level of digitalization within enterprises influences their environmental performance.

Digital transformation facilitates the development of green core capabilities and the transmission of green pressures from stakeholders, encompassing advancements in green innovation, management, and governance, which are essential for enhancing environmental performance of enterprises. This study posits four underlying mechanisms through which digital transformation improves environmental performance: the green innovation channel, the environmental monitoring and communication channel, the environmental information transparency channel, and the external attention channel.

## 1. The Green Innovation Channel

Green innovation is recognized as one of the most important factors in achieving both growth and environmental sustainability (Singh et al. 2020, Stern and Valero 2021), with digital technologies and infrastructure playing a crucial role in fostering environmental innovations in green technologies, products, processes, and services. Green products incorporate energy minimization, materials reduction, and pollution prevention into their design (Dangelico and Pujari 2010). Green processes involve more efficient production systems, the use of green manufacturing equipment and energy, as well as “end-of-pipe” control technologies and preventative measures such as carbon capture, utilization, and storage.

Innovation literature emphasizes the vital role of technological opportunities and knowledge spillovers among agents and firms in enhancing innovative capabilities (Jaffe et al. 1993, Audretsch and Feldman 1996). Drawing on resource-based theory and Schumpeter’s innovation theory, several studies empirically confirm the positive impact of corporate digital transformation on green innovation (Tang et al. 2021, Tang et al. 2023). The ability to acquire, access, and integrate declarative (know-what) and procedural (know-how) knowledge from external sources—such as customers, competitors, universities, or consultants—is critical to a firm’s innovative success (Borgatti and Cross 2003, Trantopoulos et al. 2017). Digital technologies, particularly big data analytics, can expedite the innovation process by facilitating the identification, access, combination, and deployment of existing knowledge to address new challenges, especially in decentralized innovation (Wu et al. 2019).

In the realm of green innovation, workforce skills in digital literacy have emerged as a key driver. Santoalha et al. (2021) analyze data from 142 European regions and find that e-skills endowment and relatedness positively predict regions’ abilities to specialize in new technological domains, particularly in green specializations. Beyond innovation resources, the implementation of innovation by R&D personnel has become more efficient, intelligent, and sustainable, thanks to digital technologies, notably digital twins (Tao et al. 2018) and virtual manufacturing. Progress is occurring not only at the individual level, but also within collective innovation paradigms. De Marchi (2012) emphasizes the importance of R&D cooperation with external partners in environmental innovation, asserting that cross-country and cross-domain collaborative innovation activities are essential for promoting a more environmentally sustainable economy.

Further, digital transformation also creates new technological opportunities in manufacturing. Frank et al. (2019) highlight Industry 4.0 technologies consisting of front-end technologies (e.g., smart manufacturing, smart products, smart supply chains, and smart working) and foundational technologies (e.g., IoT, cloud services, big data, and analytics). In smart manufacturing, industrial robots contribute to productivity growth and reduced output prices (Graetz and Michaels 2018), while simultaneously decreasing waste resource and promoting green technology innovation (Lee et al. 2022). Other digital processes—such as automation, smart factories, and on-demand production—enhance precise control and resource utilization, ultimately contributing to improved environmental performance. Therefore, we propose the second hypothesis as follows:

**Hypothesis 2.** *Corporate digital transformation promotes green innovation, thus enhancing environmental performance.*

## 2. The Environmental Monitoring and Communication Channel

Environmental monitoring and communication serve as foundational prerequisites for effective environmental management, playing a crucial role in enhancing management efficiency. Environmental management encompasses all efforts to minimize the negative environmental

impacts of a firm's value-creation processes, including continuous monitoring of process discharges, employee training, and environmental audits. Effective environmental management can enhance both environmental and financial performance (Klassen and McLaughlin 1996), with stronger effects observed under formal environmental management systems (Melnik et al. 2003) and green information systems (Gholami et al. 2016).

Digital transformation enhances a firm's capabilities in environmental monitoring and communication, thereby spanning company-wide business strategies, including both operational and functional strategies. From a strategic perspective, powerful information systems enable organizations to engage in sense-making regarding emerging environmental requirements and consumer demands (Seidel et al. 2013, Setia et al. 2013), as well as climate risks and uncertainties in environmental policy. This allows entrepreneurs to implement environmentally sustainable practices more effectively, facilitating a more effective response to sudden environmental events. With accurate and timely life-cycle information combined with analytical techniques, digital transformation empowers managers to make more precise strategic predictions and informed decisions, facilitating quicker responses to regulatory changes and environmental emergencies (Chen and Jiang 2024).

Specifically, digital technologies enhance environmental governance both within firms and across supply chains. Sirmon et al. (2011) emphasize the crucial role of managers in effectively structuring, bundling, and leveraging firm resources—referred to as “resource orchestration”—to create competitive advantages. For example, Benzidia et al. (2021) demonstrate, using data from French hospitals, that big data analytics and AI facilitate environmental process integration and green supply chain collaboration, ultimately improving environmental performance.

Moreover, the shift towards automation and robot adoption influences human capital management. On the one hand, digital transformation enables firms to analyze and optimize employee efficiency, necessitating improvements in workforce skills and enhancing labor productivity. It also substitutes redundant labor, creates new tasks, optimizes talent structures, and reduces labor costs (Acemoglu and Restrepo 2018, Graetz and Michaels 2018). On the other hand, the trend toward remote work and virtual or hybrid conferences, accelerated by the coronavirus disease (COVID-19) pandemic, has become more prevalent. Considering factors like food, accommodation, preparation, execution, communication, and transportation, off-site work patterns and online collaboration platforms have been shown to significantly reduce carbon footprints and energy consumption (Tao et al. 2021). Hence, the third hypothesis is proposed as follows:

**Hypothesis 3.** *Corporate digital transformation strengthens environmental monitoring and communication, thereby enhancing environmental performance.*

### **3. The Environmental Information Transparency Channel**

The information asymmetry problem poses a significant challenge to achieving efficient internal and external environmental governance, primarily stemming from a lack of data accumulation or difficulties in acquiring accurate information. In this context, digital transformation can enhance the quality of environmental information and alleviate the challenges associated with obtaining life-cycle “ecological footprints” of products and processes. Beyond improving information collection and storage, Che et al. (2023) argue that digital transformation facilitates cross-departmental collaboration, enhances organizational transparency, and supports information disclosure.

In addition to public information, analysts and news media are likely to seek out and acquire private information (Altschuler et al. 2015). Social media platforms like Facebook and Twitter have

emerged as popular channels for CSR communication, bridging firms and stakeholders (Vogler and Eisenegger 2021). Wang et al. (2023) demonstrate that the efficiency and cost of information transmission significantly affect firms' disclosure behaviors and the extent of information filtering by regulatory and market mechanisms.

Regarding the impacts of environmental disclosure on corporate environmental performance, previous empirical evidence presents mixed results because of two competing theories. The economics-based voluntary disclosure theory posits a positive association, primarily driven by superior performers' intentions to signal their commitment to sustainability (e.g., Verrecchia 1983, Clarkson et al. 2008, Shi et al. 2021). In contrast, socio-political theories predict a negative association, emphasizing that inferior performers may engage in self-serving disclosures under "legitimization" pressure (Patten 2002).

While disclosures are still evolving, improved transparency can motivate both high and low performers to enhance their environmental performance. The "monitoring effect" from institutional investors (Boone and White 2015, Abramova et al. 2020), analysts (Irani and Oesch 2013), citizen participation (Buntaine et al. 2024), and government regulation (Buntaine et al. 2024) can drive more effective environmental disclosures and actions. The "reputation effect" encourages firms to behave responsibly and maintain a positive green image because of concerns about reputation (Cao et al. 2012).

Additionally, Dhaliwal et al. (2011) argue that voluntary disclosure of CSR activities can reduce a firm's cost of equity capital, attract dedicated institutional investors, and increase analyst coverage, leading to more accurate and converging forecasts. This financing channel may also support further green investments, ultimately resulting in improved corporate environmental performance. Therefore, the fourth hypothesis is articulated as:

**Hypothesis 4.** *Corporate digital transformation increases environmental information transparency, thereby enhancing environmental performance.*

#### **4. The External Attention Channel**

As information distribution centers and public opinion platforms become increasingly active and developed, the internet provides a wide range of channels for information seeking and expression for the public and social organizations. The rise of AI, particularly through the use of large language models, has enhanced the efficiency of public information retrieval and the rapid analysis of extensive datasets. The prevalence of algorithmic recommendations has further altered attention patterns across different stakeholders. Importantly, the quality and quantity of public and private information accessible to audiences regarding enterprises are closely tied to the level of corporate digitalization.

The role of external attention in corporate governance and performance is well documented in the literature. Attention from stakeholders, such as local governments (ElMassah and Mohieldin 2020), analysts (Altschuler et al. 2015), and institutional investors (Abramova et al. 2020), reflects the intensity of governance efforts and subsequently influences corporate financial and environmental performance. Recent studies further emphasize the critical function of public citizen participation (Buntaine et al. 2024) in supervision on environmental violation activities. Online public opinion, characterized by rapid dissemination, broad reach, and strong interactivity, has increased pressure on enterprises. Therefore, greater external oversight intensity promotes heightened corporate commitment to improving environmental performance (Tang and Tang 2016), which helps secure public recognition and enhance corporate legitimacy.

Moreover, the rise of ESG investment has led investors to factor in the green investment information of listed companies in their decision-making. When corporate environmental violations are disclosed by the media and public opinion, a company's green goodwill suffers, prompting investors to reduce their investments and express negative sentiments. Particularly, firms that experience significant environmental accidents often face government penalties (such as fines, compensation, and litigation), raising investor concerns about profitability and business prospects, which can adversely affect their legal status and market value. Consequently, digital transformation plays a pivotal role in amplifying the impact of external attention on corporate environmental behavior. This leads to our fifth hypothesis:

**Hypothesis 5.** *Corporate digital transformation fosters greater external attention, thereby enhancing environmental performance.*

### III. METHOD AND DATA

#### A. Variable Measurement

##### 1. Measuring Corporate Digital Transformation

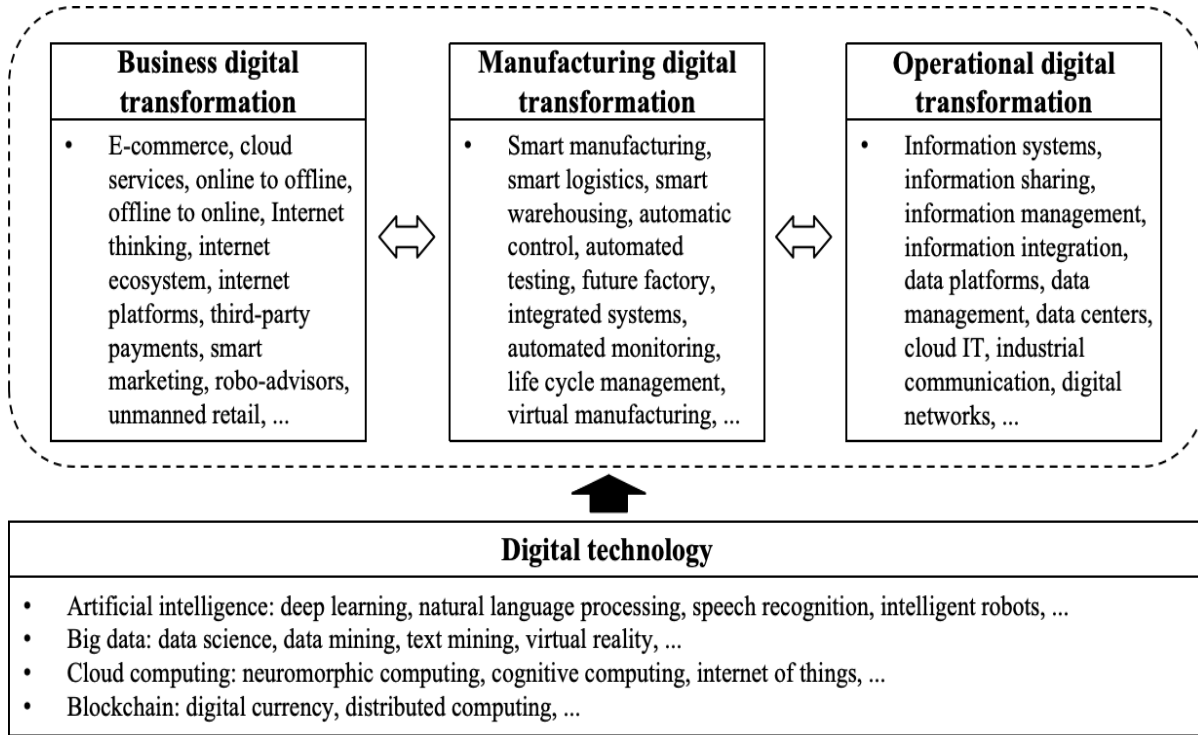
Currently, digital transformation lacks unified measurement indicators at the firm level. Building on existing literature (e.g., Du et al. 2023, Song et al. 2024), we utilize the frequency of terms related to digital transformation in firms' annual reports as an indicator of corporate digital transformation levels (*LnDT*). The technical implementation follows three specific steps.

First, we construct a digital transformation keyword dictionary. Drawing inspiration from Verhoef et al. (2021), we focus on "the application of digital technologies in enterprises" and develop a four-dimensional digital transformation measure comprising: digital technologies (*LnDT\_Technology*), business digital transformation (*LnDT\_Business*), manufacturing digital transformation (*LnDT\_Manufacturing*), and operational digital transformation (*LnDT\_Operation*). This dictionary encompasses a total of 160 keywords, as illustrated in Figure 2.

Second, we create a clean text database of annual reports from listed companies. We extract the original annual reports of A-share listed companies from 2000 to 2022 via web scraping from the CNINFO website using Python. Subsequently, we employ the Java PDFBox library to convert the initial PDF files into a text format. Building on this, we utilize Python's Jieba Chinese word segmentation library to segment the texts into sentences and perform text cleaning, which includes removing stop words (e.g., adverbs, prepositions, and conjunctions) and eliminating negative expressions such as "no," and "without," preceding the keywords.

Finally, leveraging our defined dictionary, we calculate keyword frequency across the four sub-dimensions and subsequently aggregate these to derive an overall measure of enterprise digital transformation. Given the right-skewed distribution of this data, we employ logarithmic transformation as the final measurement metric.

**Figure 2: Structured Keyword Dictionary of Corporate Digital Transformation**



Notes: This figure illustrates the sub-dimensions associated with corporate digital transformation and their corresponding common keywords. We classify “the application of digital technologies in enterprises” into four categories: digital technologies, business digital transformation, manufacturing digital transformation, and operational digital transformation. The construction of these measures is based on a textual analysis of firms’ annual reports (Section III.A.1). The sub-dimensions are derived using a keyword dictionary tailored to each sub-category.

Source: Authors’ compilation.

## 2. Measuring Corporate Environmental Performance

Existing research primarily uses comprehensive indices, pollution emissions, and ecological benefit methods to measure corporate environmental performance. Given that ESG has become a crucial indicator for the international community to assess the level of green and sustainable development in enterprises, this paper selects one key component of ESG ratings—the environmental rating score (*EScore*)—as a proxy for environmental performance. Following Lin et al. (2021), we utilize the *EScore* developed by the Sino-Securities Index (SNSI) Information Service Co. Ltd., a leading data provider specializing in ESG evaluation and green finance information in the PRC. SNSI has released ESG ratings since 2009, and its coverage of Chinese-listed firms is the largest to date.

Considering the ongoing debate regarding the divergence in ESG ratings, we employ two alternative measures of environmental performance in our robustness checks. The first measure, *EScore\_wind*, utilizes the *EScore* from the Wind ESG ratings, which has been available since 2018 and offers extensive coverage of Chinese-listed firms, thereby aligning well with the objectives of this study. The second measure, *EnvTax*, is based on environmental protection tax data disclosed in annual reports. Because of significant disparities in tax payments across different industries, direct comparisons of absolute values are inappropriate. Thus, we define



*EnvTax* based on the quartile ranking of environmental protection tax (scaled by total assets) within the same industry and year, categorizing firms from 1 to 4, where a lower score indicates a reduced ratio of environmental protection tax paid. Although the dataset is relatively small, it offers a more objective assessment.

### 3. Measuring Mediating Variables

This study examines the four plausible channels: green innovation, environmental monitoring and communication, environmental information transparency, and external attention. Measuring these channels presents challenges in our context, as not all are readily observable or quantifiable with the available data. Therefore, we utilize the most representative or accessible indicators to measure the mechanisms, aiming to present suggestive evidence that may enhance our understanding of these channels.

**Green innovation (*GPatent*).** Following existing literature (e.g., Tang et al. 2021), we characterize use the number of green patents filed (and eventually granted) by enterprises as the proxy for corporate green innovation. Because of the right-skewed distribution of this data, we employ logarithmic transformation as the measurement metric.

**Environmental monitoring and communication (*EnvReponse*).** To better address climate risks, particularly unforeseen environmental events, firms generally need to establish an environmental emergency response system. This study constructs an indicator based on whether firms disclose in their annual reports the establishment of such a response mechanism and the formulation of associated regulations and measures. A value of 1 is assigned if the firm reports this, and 0 if not.

**Environmental information transparency (*EnvDisclosure*).** In the sample period, corporate environmental information disclosure is primarily voluntary in the Chinese A-share market, resulting in significant variations in the extent of such disclosures among firms. We measure this transparency by assessing whether companies disclose environmental information in their public reports, including annual reports, CSR reports, ESG reports, or environmental reports. A value of 1 is assigned if relevant information is disclosed; otherwise, the value is 0.

**External attention (*AnaCoverage*).** Following Irani and Oesch (2013), we use analyst coverage as a proxy for external attention. This measure is constructed using the natural logarithm of one plus the number of analysts (or teams) that have conducted tracking analysis on the company during the fiscal year.

### 4. Other Variables

We follow the existing literature and control for some firm characteristics that may affect corporate environmental performance. The control variables include *LnMktCap* (natural logarithm of the annual circulating market capitalization), *ROE* (return on equity, net profit divided by average balance of shareholders' equity), *Leverage* (total liabilities divided by total assets), and *LnListAge* (natural logarithm of one plus the difference between the fiscal year and the listing year) as control variables in all regression analyses. In the heterogeneity analysis, we employ urban GDP as a measure of regional economic development.

## B. Research Methodology

To assess the impact of digital transformation on environmental performance at the firm level, we construct the baseline Ordinary-Least Squares (OLS) regression model:

$$EScore_{i,t} = \alpha_0 + \alpha_1 LnDT_{i,t-1} + \alpha_j Controls_{i,t} + \lambda_i + \lambda_t + \varepsilon_{i,t} \quad (1)$$

where  $EScore_{i,t}$  is the environmental score derived from the Sino-Securities Index ESG ratings for firm  $i$  in year  $t$ , while  $LnDT_{i,t-1}$  denotes the digital transformation level for firm  $i$  in the preceding year  $t-1$ .  $Controls$  are a vector of firm characteristics, including firm size, ROE, leverage and listing age.  $\lambda_i$  is the firm fixed effects, accounting for permanent differences in firms' environmental performance, whereas  $\lambda_t$  is the year fixed effects to control for time-varying common shocks.  $\varepsilon_{i,t}$  is the standard errors clustered at the firm level. We apply a one-period lag to the core explanatory variable  $LnDT$  to account for inherent time delays in the transmission of effects, while also addressing potential endogeneity issues related to reverse causality.

To further verify the mediating effects of the four potential channels, this study employs the mediating effect model (Mackinnon et al. 2000, Wen and Ye 2014). Specifically, to examine the indirect influence of the independent variable ( $LnDT$ ) on the dependent variable ( $EScore$ ) through the mediating variable ( $Med$ ), the following equations are utilized:

$$Med_{i,t-1} = \beta_0 + \beta_1 LnDT_{i,t-1} + \beta_j Controls_{i,t} + \lambda_i + \lambda_t + \varepsilon_{i,t} \quad (2)$$

$$EScore_{i,t} = \gamma_0 + \gamma_1 LnDT_{i,t-1} + \gamma_2 Med_{i,t-1} + \gamma_j Controls_{i,t} + \lambda_i + \lambda_t + \varepsilon_{i,t} \quad (3)$$

where  $Med$  represents  $GPatent$  (green patents),  $EnvReponse$  (the establishment of the environmental emergency response system),  $EnvDisclosure$  (the disclosure of environmental information in public reports), and  $AnaCoverage$  (analyst coverage). Since patent data and analyst coverage are count data, we use Poisson regression, as outlined by Cohn et al. (2022), when regressing these two mediating variables as the dependent variable in the equation (2). Further, in response to certain academic critiques of this method, we also incorporate an interaction term model as complementary evidence.

## C. Data and Sample

This paper selects the PRC's A-share listed companies as the sample for analysis. Data is sourced from the Sino-Securities/Wind ESG rating for information on the ESG ratings of listed firms, as well as from the China Stock Market & Accounting Research databases for other relevant information. Observations with missing data are excluded, and continuous variables are winsorized at 1%. Our final sample includes 42,008 firm-year observations spanning 5,200 companies from 2009 to 2022.

Table 1 presents the summary statistics for the variables utilized in this study. The average environmental performance ( $EScore$ ) of Chinese-listed firms, measured on a scale of 100, is 60.303, with a maximum score of 80.150 and a minimum score of 45.690, resulting in a standard deviation of 7.42. These findings underscore substantial disparities in environmental performance levels among listed companies in the PRC.

Based on our four-dimensional digital transformation dictionary, the average digital transformation level (*LnDT*) of our sample is 2.890, with a standard deviation of 1.297, indicating significant variation in the degree of digitalization among firms. Among the four dimensions, business digital transformation exhibits the highest level, followed by operational, manufacturing, and technological digital transformation, with means of 1.784, 1.592, 1.565, and 0.894, respectively.

On average, Chinese-listed firms generate 11 green patents annually, exhibiting considerable disparity among them. Only 18.5% of the sample has implemented environmental emergency response systems based on voluntary disclosure, not compliance rates, while 82.8% of firms disclose environmental information. The average analyst coverage for a listed firm in the PRC comprises 10 analyst teams. The statistics for the remaining variables fall within a reasonable range and will not be elaborated upon here.

**Table 1: Summary Statistics**

|                                                  | Obs.   | Mean   | SD     | Min    | P25    | Median | P75    | Max     |
|--------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|---------|
| <b>Corporate Digital Transformation Measures</b> |        |        |        |        |        |        |        |         |
| <i>LnDT</i>                                      | 45,460 | 2.890  | 1.297  | 0.000  | 1.946  | 2.833  | 3.784  | 5.908   |
| <i>LnDT_Technology</i>                           | 45,460 | 0.894  | 1.242  | 0.000  | 0.000  | 0.000  | 1.609  | 4.804   |
| <i>LnDT_Business</i>                             | 45,460 | 1.784  | 1.344  | 0.000  | 0.693  | 1.609  | 2.708  | 5.147   |
| <i>LnDT_Manufacturing</i>                        | 45,460 | 1.565  | 1.216  | 0.000  | 0.693  | 1.609  | 2.485  | 4.564   |
| <i>LnDT_Operation</i>                            | 45,460 | 1.592  | 1.021  | 0.000  | 0.693  | 1.609  | 2.303  | 4.290   |
| <b>Environmental Performance Measures</b>        |        |        |        |        |        |        |        |         |
| <i>EScore</i>                                    | 42,008 | 60.303 | 7.420  | 45.690 | 55.080 | 60.090 | 65.190 | 80.150  |
| <i>EScore_wind</i>                               | 19,911 | 1.649  | 2.003  | 0.000  | 0.000  | 1.010  | 2.470  | 8.780   |
| <i>EnvTax</i>                                    | 4,599  | 0.358  | 0.774  | 0.000  | 0.009  | 0.056  | 0.299  | 4.769   |
| <b>Mediating Variables</b>                       |        |        |        |        |        |        |        |         |
| <i>GPatent</i>                                   | 12,707 | 10.595 | 23.594 | 1.000  | 1.000  | 3.000  | 9.000  | 179.000 |
| <i>EnvResponse</i>                               | 42,270 | 0.185  | 0.388  | 0.000  | 0.000  | 0.000  | 0.000  | 1.000   |
| <i>EnvDisclosure</i>                             | 40,508 | 0.828  | 0.377  | 0.000  | 1.000  | 1.000  | 1.000  | 1.000   |
| <i>AnaCoverage</i>                               | 30,842 | 9.947  | 10.090 | 1.000  | 2.000  | 6.000  | 14.000 | 46.000  |
| <b>Controls</b>                                  |        |        |        |        |        |        |        |         |
| <i>LnMktCap</i>                                  | 45,460 | 15.216 | 1.176  | 12.797 | 14.412 | 15.121 | 15.891 | 18.745  |
| <i>ROE</i>                                       | 45,064 | 0.058  | 0.159  | -0.872 | 0.028  | 0.075  | 0.126  | 0.388   |
| <i>Leverage</i>                                  | 45,459 | 0.438  | 0.223  | 0.050  | 0.260  | 0.425  | 0.598  | 0.985   |
| <i>LnListAge</i>                                 | 45,460 | 2.032  | 0.948  | 0.000  | 1.386  | 2.303  | 2.833  | 3.332   |

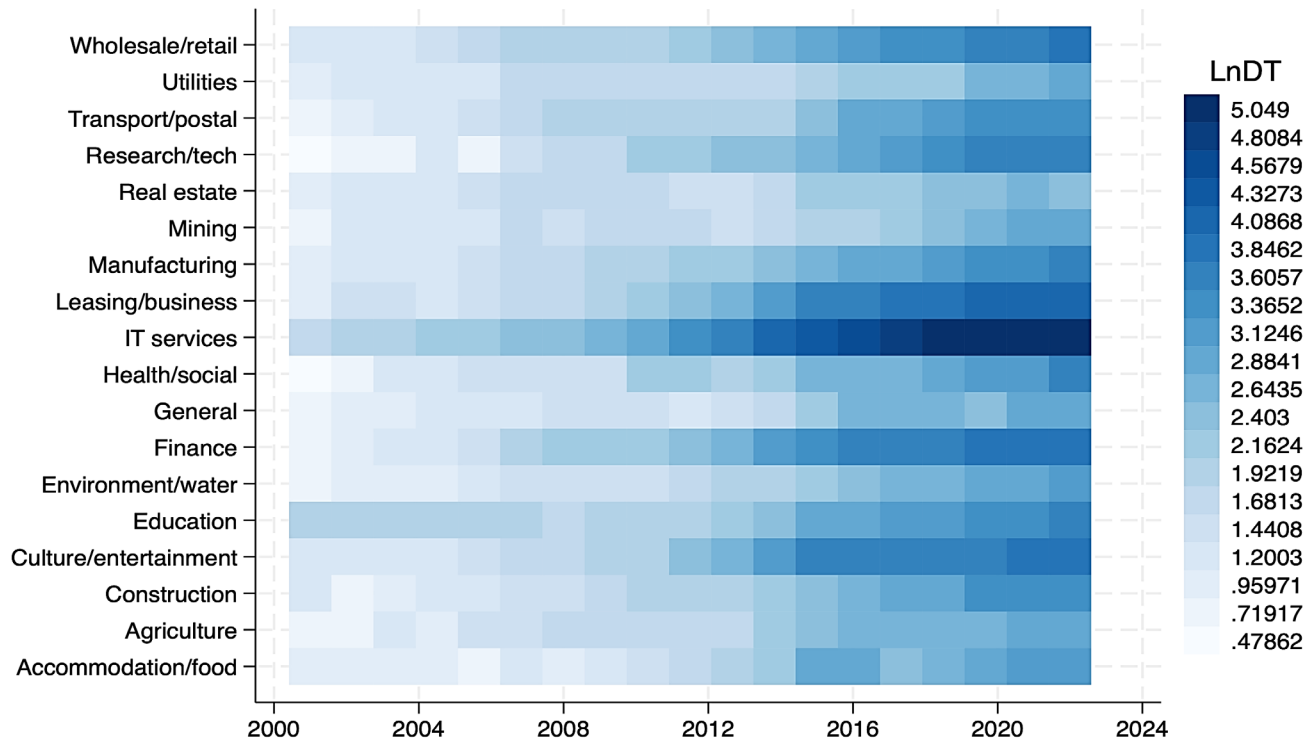
ROE = return on equity,

Notes: This table presents the summary statistics for the major variables of this study. After excluding observations with missing data, the final sample includes 42,008 firm-year observations from 5,200 companies between 2009 and 2022. The continuous variables are winsorized at 1%. Detailed definitions of the variables are provided in Appendix 1. The correlations between the key variables are reported in Appendix 2.

Source: Authors' calculations.

Figure 3 illustrates the yearly average digital transformation levels across industries, with firms categorized into 18 sectors according to the 2012 CSRC Industry Classification. From a temporal perspective, digitalization levels across industries have generally increased year by year during the sample period. In a cross-industry comparison, with the exception of the IT services sector, industries such as leasing and business services, wholesale and retail, financial services, and entertainment have led in digital transformation. Conversely, sectors like real estate, utilities, mining, and agriculture have shown slow progress, while the manufacturing sector has accelerated its transformation over the past decade.

**Figure 3: Average Overall Digital Transformation Levels Across Sectors**



Notes: This figure illustrates the yearly average overall digital transformation levels ( $LnDT$ ) across sectors from 2000 to 2023. The construction of these measures is based on a textual analysis of firms' annual reports (Section III.A.1). Firms are categorized into 18 sectors according to the 2012 Industry Classification by the China Securities Regulatory Commission.

Source: Authors' compilation.

#### IV. EMPIRICAL RESULTS

In this section, we first present the baseline results and address endogeneity concerns to establish the causal relationship between digital transformation and environmental performance, subsequently validating the robustness of the regression findings. We then report the results of mechanism tests and a relative importance analysis of the four proposed channels. Additionally, we conduct a heterogeneity analysis and further examine the impact of the various sub-dimensions of our digital transformation measure.

## A. Benchmark Regression

In the baseline model, we examine the direct effects of corporate digital transformation on environmental performance. Table 2 presents the regression results for equation (1). Column (1) evaluates the net impact of digital transformation on environmental performance without controlling for firm-level characteristics for comparative purposes, while column (2) includes the control variables. Both models include year fixed effects and firm fixed effects, with standard errors clustered at the firm level.

The OLS regression results show that the coefficient estimates for  $LnDT$  are both positive and statistically significant, suggesting a substantial positive effect of corporate digital transformation on environmental performance. The economic impact is notable: ceteris paribus, a one-standard-deviation increase in corporate digital transformation from its mean results in a 1.34% improvement in corporate  $EScore$  in the subsequent year. Because of the limited coverage of our digital transformation dictionary in capturing all relevant digital transformation terminology, this economic benefit may be still underestimated. This finding indicates that firms' efforts in digital transformation will help enhance their environmental performance, validating the effectiveness of digitalization and environmental sustainability at the firm level.

Regarding the control variables, we find that the impacts of firm size and listing age on environmental performance are significantly positive at the 1% significance level. However, ROE and leverage are not significantly associated with the  $EScore$  of Chinese-listed firms. This may reflect that, compared to profitability and financial status, corporate ecological responsiveness is more influenced by stakeholder pressures and leadership responsibility toward the environment. More mature firms often serve as role models in sustainability practices.

**Table 2: Corporate Digital Transformation and Environmental Performance**

|                    | $EScore_{i,t}$       |                      |
|--------------------|----------------------|----------------------|
|                    | (1)                  | (2)                  |
| $LnDT_{i,t-1}$     | 0.672***<br>(0.065)  | 0.621***<br>(0.064)  |
| $LnMktCap_{i,t}$   |                      | 0.332***<br>(0.085)  |
| $ROE_{i,t}$        |                      | -0.384<br>(0.245)    |
| $Leverage_{i,t}$   |                      | -0.304<br>(0.365)    |
| $LnListAge_{i,t}$  |                      | 1.001***<br>(0.193)  |
| Constant           | 58.297***<br>(0.181) | 51.275***<br>(1.229) |
| Firm fixed effects | Yes                  | Yes                  |
| Year fixed effects | Yes                  | Yes                  |
| Observations       | 38,524               | 38,249               |
| $R^2$              | 0.633                | 0.635                |

ESG = environmental, social, and governance; ROE = return on equity,

Notes: This table presents the baseline results from a firm-year level estimation, regressing corporate digital transformation ( $LnDT$ ) on environmental performance ( $EScore$ ).  $EScore$  represents firms' annual environmental scores from the Sino-Securities Index ESG Rating system. The core independent variable,  $LnDT$ , is the natural logarithm of the corporate digital transformation level plus one for year  $t-1$ , derived from textual analysis of firms' annual reports. Control variables include firm size ( $LnMktCap$ ), return on equity ( $ROE$ ), leverage ( $Leverage$ ), and listing age ( $LnListAge$ ). All variables are defined in Appendix 1. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

## B. Endogenous Tests

The endogeneity of a model mainly originates from two sources: omitted variable bias and reciprocal causation. For the omitted variable issue, we have incorporated both year fixed effects and firm fixed effects, along with relevant control variables, in the baseline model. In practice, digital transformation and environmental performance may be concurrently influenced by certain unobserved factors, which increases the potential for reciprocal causation. In the baseline model, we have lagged the core independent variable by one period; to further mitigate this concern, we use an instrumental variable approach and conduct a DID analysis based on an exogenous policy shock.

A suitable instrumental variable must not only be correlated with the endogenous variable, but also satisfy the exogeneity condition (Angrist and Pischke 2009). This study uses two instrumental variables based on peer firms' digital transformation levels. The first instrumental variable (*IV\_peer*) is defined as the average digital transformation level of other firms within the same industry. Given the skewness of the data, we also follow Lewbel (1997) and employ the cubic difference between a firm's digital transformation level and the industry average digitalization level as the second instrumental variable (*IV\_Lewbel*). As shown in columns (1) and (3) of Table 3, the first-stage regression results indicate that the coefficient of the instrumental variable for digital transformation is significantly positive, suggesting the existence of peer effects and potential spillover effects at the industry level. Moreover, the Sanderson-Windmeijer F-statistic satisfies the correlation requirement for the instrumental variable. The Kleibergen-Paap LM statistic rejects the null hypothesis of "under-identified instruments." Meanwhile, the Cragg-Donald Wald F-statistic and the Kleibergen-Paap rk Wald F-statistic for weak identification exceed the critical threshold at the 1% significance level. In the second-stage results, presented in columns (2) and (4), the coefficients remain significantly positive.

Following the existing literature (e.g., Tang et al. 2021, Wang et al. 2022), we take the "Broadband China" strategy as a quasi-natural experiment and apply the DID method to further demonstrate the causal relationship between digital transformation and environmental performance. The "Broadband China" strategy, issued by the State Council of the PRC in August 2013, marked the official initiation of telecommunications infrastructure as a national priority, with 120 cities designated as demonstration cities between 2014 and 2016. Higher internet penetration indicates a more conducive environment for the development of digital technologies by firms. Column (5) of Table 4 presents the staggered DID regression results, which suggest that improvements in digital infrastructure at the city level tend to affect the industrial structure and operational modes of entire cities, thereby benefiting enterprises' digitalization and ultimately enhancing their environmental performance.

## C. Robustness Tests

We conduct a series of robustness tests on our baseline results, including the use of alternative environmental performance measures; a change-on-change specification; the incorporation of concurrent or lagged digital transformation; and improving sample quality by excluding financial firms, information services, and special treatment-listed firms. These checks are designed to verify the robustness of our findings, and are reported in Table 4.

Columns (1) and (2) in Panel A present the results for the two alternative environmental performance measures: the EScore from the Wind ESG ratings (*EScore\_wind*) and the environmental protection tax (*EnvTax*). The former addresses concerns regarding the divergence in ESG ratings, while the latter serves as a more objective measure of environmental performance

that also incorporates economic implications. Column (1) confirms a significant positive relationship between digital transformation and environmental performance. In Column (2), the coefficient of lagged  $LnDT$  on  $EnvTax$  is significantly negative at the 10% level, indicating that digital transformation enables firms to reduce their environmental tax payment intensity compared to their industry peers.

**Table 3: Endogeneity Tests**

|                                     | $LnDT_{i,t}$        | $EScore_{i,t}$      | $LnDT_{i,t}$        | $EScore_{i,t}$      | $EScore_{i,t}$     |
|-------------------------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
|                                     | (1)                 | (2)                 | (3)                 | (4)                 | (5)                |
| $IV\_peer_{i,t}$                    | 0.717***<br>(0.054) |                     |                     |                     |                    |
| $IV\_Lewbel_{i,t}$                  |                     |                     | 0.121***<br>(0.004) |                     |                    |
| $LnDT_{i,t-1}$                      |                     | 1.536***<br>(0.487) |                     | 0.534***<br>(0.094) |                    |
| $Treat \times Post$                 |                     |                     |                     |                     | 0.374**<br>(0.166) |
| Control variables                   | Yes                 | Yes                 | Yes                 | Yes                 | Yes                |
| Firm fixed effects                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                |
| Year fixed effects                  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                |
| Sanderson-Windmeijer F test         | 175.22<br>[0.000]   |                     | 808.86<br>[0.000]   |                     |                    |
| Kleibergen-Paap rk LM statistic     |                     | 135.643<br>[0.000]  |                     | 659.916<br>[0.000]  |                    |
| Cragg-Donald Wald F statistic       |                     | 945.075             |                     | 2.9e+04             |                    |
| Kleibergen-Paap rk Wald F statistic |                     | 175.222             |                     | 808.864             |                    |
| Observations                        | 38,245              | 38,245              | 38,249              | 38,249              | 41,314             |

Note: This table presents two sets of tests addressing potential endogeneity bias. The first set uses instrumental variables (IV) regressions. In columns (1) and (2), the instrumental variable  $IV\_peer$  represents the average digital transformation level of peer firms within the same industry and fiscal year. In columns (3) and (4), the instrumental variable  $IV\_Lewbel$ , following Lewbel (1997), is defined as the cubic deviation of a firm's digital transformation level from the industry average in the same fiscal year. The second set employs a staggered difference-in-differences (DID) approach, using the "Broadband China" strategy as a shock, which was rolled out in batches across 120 cities from 2014 to 2016. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

In the preceding empirical analysis, we utilized the end-of-period status of enterprises as the core variable. Considering the "path dependence" of both digitalization and sustainability progress, we further investigate the relationship between annual changes in digital transformation and environmental performance to explore their dynamic effects. Specifically, we use  $\Delta LnDT$  as the independent variable and  $\Delta lnEScore$  as the dependent variable in our model. As shown in column (3) in Panel A, higher increments in digital transformation are associated with greater increases in EScore. Interestingly, we observe a significant negative correlation between changes in digital transformation and changes in environmental performance within the same period, as indicated in column (4) of Panel A. This suggests that, in the short term, the improvements in EScore

resulting from increased digital transformation investments may be offset by the higher energy consumption and pollution associated with those efforts. This result not only confirms the economic rationale and robustness of the baseline model, but also provides insights into the dynamic relationship between digital transformation and environmental performance.

**Table 4: Robustness Tests**

| <b>Panel A: Alternative Environmental Performance Measures and Change-on-Change Model Specification</b> |                                  |                             |                                |                      |
|---------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------|--------------------------------|----------------------|
|                                                                                                         | <i>EScore_wind<sub>i,t</sub></i> | <i>EnvTax<sub>i,t</sub></i> | <i>ΔLnEScore<sub>i,t</sub></i> |                      |
|                                                                                                         | (1)                              | (2)                         | (3)                            | (4)                  |
| <i>LnDT<sub>i,t-1</sub></i>                                                                             | 0.102***<br>(0.027)              | -0.053*<br>(0.029)          |                                |                      |
| <i>ΔLnDT<sub>i,t-1</sub></i>                                                                            |                                  |                             | 0.010***<br>(0.001)            |                      |
| <i>ΔLnDT<sub>i,t</sub></i>                                                                              |                                  |                             |                                | -0.003***<br>(0.001) |
| Constant                                                                                                | -0.165<br>(0.442)                | 2.893***<br>(0.450)         | 0.004<br>(0.015)               | -0.113***<br>(0.015) |
| Control variables                                                                                       | Yes                              | Yes                         | Yes                            | Yes                  |
| Firm fixed effects                                                                                      | Yes                              | Yes                         | Yes                            | Yes                  |
| Year fixed effects                                                                                      | Yes                              | Yes                         | Yes                            | Yes                  |
| Observations                                                                                            | 18,362                           | 3,917                       | 34,278                         | 36,280               |
| R <sup>2</sup>                                                                                          | 0.751                            | 0.843                       | 0.071                          | 0.078                |

| <b>Panel B: Replacing Digital Transformation Measures</b> |                             |                      |                      |
|-----------------------------------------------------------|-----------------------------|----------------------|----------------------|
|                                                           | <i>EScore<sub>i,t</sub></i> |                      |                      |
|                                                           | (1)                         | (2)                  | (3)                  |
| <i>LnDT<sub>i,t</sub></i>                                 | 0.273***<br>(0.061)         |                      |                      |
| <i>LnDT<sub>i,t-2</sub></i>                               |                             | 0.385***<br>(0.061)  |                      |
| <i>LnDT<sub>i,t-3</sub></i>                               |                             |                      | 0.221***<br>(0.061)  |
| Constant                                                  | 55.501***<br>(1.176)        | 51.293***<br>(1.312) | 51.426***<br>(1.659) |
| Control variables                                         | Yes                         | Yes                  | Yes                  |
| Firm fixed effects                                        | Yes                         | Yes                  | Yes                  |
| Year fixed effects                                        | Yes                         | Yes                  | Yes                  |
| Observations                                              | 41,238                      | 35,193               | 32,614               |
| R <sup>2</sup>                                            | 0.608                       | 0.637                | 0.639                |

| <b>Panel C: Sample Range Adjustments</b> |                                         |                                                 |                                                  |                                   |
|------------------------------------------|-----------------------------------------|-------------------------------------------------|--------------------------------------------------|-----------------------------------|
|                                          | <i>EScore<sub>i,t</sub></i>             |                                                 |                                                  |                                   |
|                                          | (1)<br>Exclude<br>Financial<br>Industry | (2)<br>Exclude Information<br>Services Industry | (3)<br>Exclude Special<br>Treatment-Listed Firms | (4)<br>Exclude All<br>Three Types |
| <i>LnDT<sub>i,t-1</sub></i>              | 0.634***<br>(0.066)                     | 0.604***<br>(0.063)                             | 0.585***<br>(0.065)                              | 0.577***<br>(0.067)               |
| Constant                                 | 50.958***<br>(1.243)                    | 50.883***<br>(1.238)                            | 52.224***<br>(1.285)                             | 51.333***<br>(1.302)              |
| Control variables                        | Yes                                     | Yes                                             | Yes                                              | Yes                               |
| Firm fixed effects                       | Yes                                     | Yes                                             | Yes                                              | Yes                               |
| Year fixed effects                       | Yes                                     | Yes                                             | Yes                                              | Yes                               |
| Observations                             | 37,117                                  | 35,415                                          | 36,676                                           | 33,016                            |
| R <sup>2</sup>                           | 0.631                                   | 0.643                                           | 0.638                                            | 0.640                             |

ESG = environmental, social, and governance.

*Continued on the next page*



Note: This table presents four sets of robustness tests. In columns (1)–(2) of Panel A, alternative environmental performance measures are used as the dependent variable. *EScore\_wind* is the environmental score from Wind ESG Rating system. *EnvTax* represents the quartile ranking of environmental protection tax (scaled by total assets) within the same industry and year, with a lower score indicating a reduced environmental tax ratio. Columns (3)–(4) of Panel A test the impact of changes in digital transformation on changes in environmental performance.  $\Delta \ln EScore$  represents the logged difference in *EScore* between the current and previous periods,  $\Delta \ln DT$  is the change in *LnDT* between the current and previous periods. Panel B replaces the independent variable with digital transformation levels for the current period, two-period, and three-period lags. In Panel C, different samples are excluded for robustness. Column (1) of Panel C excludes the finance sector (industry J codes in the financial industry), column (2) of Panel C excludes the information services industry (industry I codes in the information transmission, software, and IT services industries), column (3) of Panel C removes special treatment-listed firms, and column (4) of Panel C excludes all three categories simultaneously. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

We then replace the core independent variable, *LnDT*, with concurrent or lagged specifications to assess whether the effect of digital transformation persists. Column (1) in Panel B shows a positive relationship between digital transformation and environmental performance within the same period. Combined with the earlier concurrent change-on-change result, this indicates that, although short-term increases in digital transformation may lead to short-term declines in environmental performance growth, overall, digital transformation still contributes to an increase in environmental performance. In columns (2) and (3) of Panel B, the coefficients of *LnDT* lagged by two and three periods are both significantly positive at the 1% level, reflecting that corporate digital transformation exerts a cumulative facilitating effect on environmental performance over time, demonstrating sustained positive feedback pathways.

To avoid potential data distortion and the overestimation of the overall level of digitalization because of certain companies' pronounced digital attributes, we exclude the following samples and reexamine our baseline results: (i) firms in the financial industry; (ii) firms in the information services industry that primarily provide information transmission, software, and IT services; and (iii) companies labeled as ST, ST\*, or PT. The regression results, after narrowing the sample range, are presented in Panel C. The coefficient size and significance level of lagged *LnDT* remain consistent with the baseline regression, indicating that our results hold reliably after the sample adjustments.

## D. Mechanism Tests

### 1. Corporate Digital Transformation, Green Innovation, and Environmental Performance

The previous theoretical analysis suggests that digital transformation enhances environmental performance by promoting green innovation. To test Hypothesis 2, we use the number of green patents filed (and eventually granted) as a proxy for firms' green technological innovation capabilities. The results are presented in Table 5. Column (2) shows a significantly positive effect of digital transformation on corporate green technological innovation output. By comparing columns (1) and (3), we demonstrate that green innovation is a key mechanism through which digital transformation improves environmental performance. The interaction term in column (4) further confirms the significant promoting effect of green patents.

One potential explanation for this finding is that digital transformation enhances the efficiency of information acquisition and dissemination, removing barriers to information flow. This facilitates timely exchange of data, information, and knowledge across the value chain, enhancing R&D personnel's capacity for green innovation. Additionally, digital transformation unlocks technological opportunities related to green products and processes, and fosters both intra- and inter-firm R&D collaboration, which collectively promote corporate green innovation, ultimately improving environmental performance. Thus, Hypothesis 2, concerning the mediating role of green innovation, is validated.

**Table 5: The Green Innovation Channel**

|                                                         | <i>EScore<sub>i,t</sub></i><br>(1) | <i>GPatent<sub>i,t</sub></i><br>(2) | <i>EScore<sub>i,t</sub></i><br>(3) | <i>EScore<sub>i,t</sub></i><br>(4) |
|---------------------------------------------------------|------------------------------------|-------------------------------------|------------------------------------|------------------------------------|
| <i>LnDT<sub>i,t-1</sub></i>                             | 0.622***<br>(0.064)                |                                     | 0.607***<br>(0.064)                | 0.518***<br>(0.068)                |
| <i>LnDT<sub>i,t</sub></i>                               |                                    | 0.085***<br>(0.027)                 |                                    |                                    |
| <i>LnGPatent<sub>i,t-1</sub></i>                        |                                    |                                     | 0.358***<br>(0.054)                | -0.139<br>(0.123)                  |
| <i>LnDT<sub>i,t-1</sub> × LnGPatent<sub>i,t-1</sub></i> |                                    |                                     |                                    | 0.166***<br>(0.038)                |
| Constant                                                | 51.391***<br>(1.228)               | -6.091***<br>(0.454)                | 51.556***<br>(1.225)               | 51.863***<br>(1.225)               |
| Control variables                                       | Yes                                | Yes                                 | Yes                                | Yes                                |
| Firm fixed effects                                      | Yes                                | Yes                                 | Yes                                | Yes                                |
| Year fixed effects                                      | Yes                                | Yes                                 | Yes                                | Yes                                |
| Observations                                            | 38,249                             | 31,089                              | 38,249                             | 38,249                             |
| R <sup>2</sup>                                          | 0.635                              | 0.302                               | 0.636                              | 0.636                              |

Notes: This table presents the regression results for testing the green innovation channel. *GPatent* is the number of green patents filed (and eventually granted) by the enterprise. Columns (1)–(3) report the results of the mediating models specified in Section II.A.2. Given that patent data are count data, Poisson regression is employed in column (2), following Cohn et al. (2022). The results remain consistent across all models using Poisson regression. Column (4) includes the interaction term between *LnDT* and *LnGPatent*. Control variables and fixed effects are aligned with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

## 2. Corporate Digital Transformation, Environmental Monitoring and Communication, and Environmental Performance

When examining whether effective environmental monitoring and communication serve as an underlying mechanism linking digital transformation to environmental performance, we acknowledge that accurately measure this channel is challenging. In practice, when comprehensive and accurate environmental monitoring, coupled with an efficient communication framework, is ensured, enterprises are better positioned to establish a system for effectively addressing environmental emergencies. Such systems may reinforce the positive relationship between digital transformation and environmental performance. Consequently, from an institutional perspective, we use the establishment of an environmental emergency response system (*EnvResponse*) as a proxy for this mechanism. The results are shown in Table 6.

Column (2) shows that the impact of digital transformation on such system is significantly positive. Comparing columns (1) and (3), we show that the ability to manage environmental emergencies partially mediates the relationship between digital transformation and environmental performance. The interaction regression result in column (4) further corroborates the significant promoting effect

of environmental monitoring and communication. In more digitized firms, managers have better access to timely and accurate information regarding resources, pollutants, labor productivity, operational processes, and external environmental events through enhanced digital transformation. Combined with improved capabilities to handle large volumes of multi-type data, they can make quicker, more rational, and well-informed decisions. As a result, firms develop stronger environmental management capabilities, in terms of internal control and external responses to unexpected environmental incidents, ultimately leading to improved environmental performance. Thus, Hypothesis 3 about the environmental monitoring and communication channel is validated.

**Table 6: The Environmental Monitoring and Communication Channel**

|                                                           | <i>EScore<sub>i,t</sub></i><br>(1) | <i>EnvResponse<sub>i,t</sub></i><br>(2) | <i>EScore<sub>i,t</sub></i><br>(3) | <i>EScore<sub>i,t</sub></i><br>(4) |
|-----------------------------------------------------------|------------------------------------|-----------------------------------------|------------------------------------|------------------------------------|
| <i>LnDT<sub>i,t-1</sub></i>                               | 0.665***<br>(0.066)                |                                         | 0.660***<br>(0.066)                | 0.624***<br>(0.069)                |
| <i>LnDT<sub>i,t</sub></i>                                 |                                    | 0.006*<br>(0.003)                       |                                    |                                    |
| <i>EnvResponse<sub>i,t-1</sub></i>                        |                                    |                                         | 1.286***<br>(0.131)                | 0.655**<br>(0.308)                 |
| <i>LnDT<sub>i,t-1</sub> × EnvResponse<sub>i,t-1</sub></i> |                                    |                                         |                                    | 0.224**<br>(0.108)                 |
| Constant                                                  | 50.625***<br>(1.278)               | -0.063<br>(0.069)                       | 50.689***<br>(1.271)               | 50.832***<br>(1.271)               |
| Control variables                                         | Yes                                | Yes                                     | Yes                                | Yes                                |
| Firm fixed effects                                        | Yes                                | Yes                                     | Yes                                | Yes                                |
| Year fixed effects                                        | Yes                                | Yes                                     | Yes                                | Yes                                |
| Observations                                              | 35,614                             | 41,395                                  | 35,614                             | 35,614                             |
| R <sup>2</sup>                                            | 0.677                              | 0.506                                   | 0.679                              | 0.679                              |

Notes: This table presents the regression results for testing the environmental monitoring and communication channel. *EnvResponse* represents the establishment of an emergency response mechanism for major environmental events within a firm, with a value of 1 if established and 0 otherwise. Columns (1)–(3) report the results of the mediating models specified in Section II.A.2. Column (4) includes the interaction term between *LnDT* and *EnvResponse*. Control variables and fixed effects are aligned with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

### 3. Corporate Digital Transformation, Environmental Information Transparency, and Environmental Performance

Table 7 presents the regression results of environmental information transparency as a mediating variable, using the discretionary disclosure of environmental information in public reports as a proxy. Column (2) indicates that digital transformation significantly promotes the disclosure of environmental information. One major reason for this improvement may be that digital transformation addresses the challenges associated with environmental data collection. Digital transformation enables firms to obtain accurate quantitative environmental information, such as energy and resource usage, carbon emissions, and pollutant discharges, thereby facilitating better information disclosure and reducing information asymmetry.

A comparison of columns (1) and (3), along with the evidence presented in column (4), illustrates that improved environmental information transparency mediates the relationship between digital transformation and environmental performance. The decision of firms to disclose relevant information primarily depends on data availability, stakeholder pressures, and leadership

responsibility. Digital transformation fosters a more transparent and informative environment, allowing green pressures to be transmitted from external to internal stakeholders. This dynamic supports and amplifies monitoring and reputation effects, ultimately leading to enhanced environmental performance.

**Table 7: The Environmental Information Transparency Channel**

|                                                             | <i>EScore<sub>i,t</sub></i><br>(1) | <i>EnvDisclosure<sub>i,t</sub></i><br>(2) | <i>EScore<sub>i,t</sub></i><br>(3) | <i>EScore<sub>i,t</sub></i><br>(4) |
|-------------------------------------------------------------|------------------------------------|-------------------------------------------|------------------------------------|------------------------------------|
| <i>LnDT<sub>i,t-1</sub></i>                                 | 0.685***<br>(0.068)                |                                           | 0.635***<br>(0.066)                | 0.297***<br>(0.086)                |
| <i>LnDT<sub>i,t</sub></i>                                   |                                    | 0.018***<br>(0.004)                       |                                    |                                    |
| <i>EnvDisclosure<sub>i,t-1</sub></i>                        |                                    |                                           | 2.418***<br>(0.103)                | 1.417***<br>(0.198)                |
| <i>LnDT<sub>i,t-1</sub> × EnvDisclosure<sub>i,t-1</sub></i> |                                    |                                           |                                    | 0.419***<br>(0.073)                |
| Constant                                                    | 51.289***<br>(1.309)               | 0.450***<br>(0.066)                       | 50.859***<br>(1.276)               | 51.794***<br>(1.273)               |
| Control variables                                           | Yes                                | Yes                                       | Yes                                | Yes                                |
| Firm fixed effects                                          | Yes                                | Yes                                       | Yes                                | Yes                                |
| Year fixed effects                                          | Yes                                | Yes                                       | Yes                                | Yes                                |
| Observations                                                | 33,930                             | 39,581                                    | 33,930                             | 33,930                             |
| R <sup>2</sup>                                              | 0.708                              | 0.397                                     | 0.717                              | 0.717                              |

Notes: This table presents the regression results for testing the environmental information transparency channel. *EnvDisclosure* represents the disclosure of relevant environmental information in public reports, assigned a value of 1 if disclosed and 0 otherwise. Columns (1)–(3) report the results of the mediating models specified in Section II.A.2. Column (4) includes the interaction term between *LnDT* and *EnvDisclosure*. Control variables and fixed effects are aligned with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

#### 4. Corporate Digital Transformation, External Attention, and Environmental Performance

Table 8 presents the test results for the external attention coverage channel, utilizing analyst coverage as a proxy. Column (2) indicates a significant positive relationship between digital transformation and analyst coverage, suggesting that firms with higher levels of digital transformation tend to receive greater exposure from external stakeholders, including analysts, government entities, media, investors, and the public. More digitized firms are associated with an increase in both the quantity and quality of available public and private information.

A comparison of columns (1) and (3), along with the evidence presented in column (4), demonstrates that greater external attention mediates the relationship between digital transformation and environmental performance. In an era of information overload and limited public attention, companies facing heightened external attention are generally more responsive to stakeholder pressures and experience improved governance. This heightened scrutiny motivates them to enhance their environmental performance.

**Table 8: The External Attention Channel**

|                                                             | <i>EScore<sub>i,t</sub></i><br>(1) | <i>AnaCoverage<sub>i,t</sub></i><br>(2) | <i>EScore<sub>i,t</sub></i><br>(3) | <i>EScore<sub>i,t</sub></i><br>(4) |
|-------------------------------------------------------------|------------------------------------|-----------------------------------------|------------------------------------|------------------------------------|
| <i>LnDT<sub>i,t-1</sub></i>                                 | 0.621***<br>(0.064)                |                                         | 0.613***<br>(0.069)                | 0.390***<br>(0.114)                |
| <i>LnDT<sub>i,t</sub></i>                                   |                                    | 0.050***<br>(0.009)                     |                                    |                                    |
| <i>LnAnaCoverage<sub>i,t-1</sub></i>                        |                                    |                                         | 0.187***<br>(0.065)                | -0.148<br>(0.145)                  |
| <i>LnDT<sub>i,t-1</sub> × LnAnaCoverage<sub>i,t-1</sub></i> |                                    |                                         |                                    | 0.114**<br>(0.046)                 |
| Constant                                                    | 51.275***<br>(1.229)               | -4.644***<br>(0.177)                    | 52.966***<br>(1.429)               | 53.744***<br>(1.441)               |
| Control variables                                           | Yes                                | Yes                                     | Yes                                | Yes                                |
| Firm fixed effects                                          | Yes                                | Yes                                     | Yes                                | Yes                                |
| Year fixed effects                                          | Yes                                | Yes                                     | Yes                                | Yes                                |
| Observations                                                | 38,249                             | 30,183                                  | 26,766                             | 26,766                             |
| R <sup>2</sup>                                              | 0.635                              | 0.506                                   | 0.667                              | 0.668                              |

Notes: This table presents the regression results for testing the external attention channel. *AnaCoverage* represents the number of analysts (teams) conducting tracking analysis on the company during the fiscal year. Columns (1)–(3) report the results of the mediating models specified in Section II.A.2. Given that analyst coverage data are count data, Poisson regression is employed in column (2), following Cohn et al. (2022). The results remain consistent across all models using Poisson regression. Column (4) includes the interaction term between *LnDT* and *LnAnaCoverage*. Control variables and fixed effects are aligned with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

## 5. Mechanism Importance Analysis

To further assess the importance of the four channels, we employ the horse race approach, standardizing all continuous variables to facilitate direct comparison of the coefficients. Table 9 presents the results of these tests. The regression includes four interaction terms between digital transformation and the four mechanism measures.

The findings indicate that, with the exception of environmental monitoring and communication, all three other channels exhibit significant positive effects through which digital transformation enhances environmental performance. The lack of significance for *EnvResponse* may be attributed to the somewhat narrow measure of the channel; nonetheless, its influence should not be overlooked. Among the three significant channels, the environmental information transparency channel is the most prominent in our context, followed by green innovation and external attention. This result not only reflects the considerable information asymmetry regarding environmental issues among Chinese-listed companies, but also underscores the necessity for both enterprises and governments to prioritize the collection, monitoring, and voluntary disclosure of environmental information as they advance the synergy between digitalization and sustainable development.

**Table 9: Mechanism Relative Importance Analysis**

|                                                                     | <i>EScore_std<sub>i,t</sub></i><br>(1) |
|---------------------------------------------------------------------|----------------------------------------|
| <i>LnDT_std<sub>i,t-1</sub></i>                                     | 0.049***<br>(0.017)                    |
| <i>LnDT_std<sub>i,t-1</sub> × LnGPatent_std<sub>i,t-1</sub></i>     | 0.024***<br>(0.007)                    |
| <i>LnDT_std<sub>i,t-1</sub> × EnvResponse<sub>i,t-1</sub></i>       | 0.024<br>(0.021)                       |
| <i>LnDT_std<sub>i,t-1</sub> × EnvDisclosure<sub>i,t-1</sub></i>     | 0.045***<br>(0.014)                    |
| <i>LnDT_std<sub>i,t-1</sub> × LnAnaCoverage_std<sub>i,t-1</sub></i> | 0.018***<br>(0.007)                    |
| <i>LnGPatent_std<sub>i,t-1</sub></i>                                | 0.029***<br>(0.007)                    |
| <i>EnvResponse<sub>i,t-1</sub></i>                                  | 0.143***<br>(0.019)                    |
| <i>EnvDisclosure<sub>i,t-1</sub></i>                                | 0.363***<br>(0.016)                    |
| <i>LnAnaCoverage_std<sub>i,t-1</sub></i>                            | 0.019**<br>(0.008)                     |
| Constant                                                            | -0.737***<br>(0.190)                   |
| Control variables                                                   | Yes                                    |
| Firm fixed effects                                                  | Yes                                    |
| Year fixed effects                                                  | Yes                                    |
| Observations                                                        | 26,766                                 |
| R <sup>2</sup>                                                      | 0.679                                  |

Notes: This table presents the regression results for the importance of the four mechanisms, analyzed using the horse race approach. To enable direct comparison of the regression coefficients, continuous key variables are standardized. The regressions include four interaction terms: *LnDT\_std* with *LnGPatent\_std*, *EnvResponse*, *EnvDisclosure*, and *LnAnaCoverage\_std*. Variable definitions are provided in Appendix 1. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors, in parentheses, are clustered by firm. \*\*\*, \*\*, and \* denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

## D. Heterogeneity Analysis

### 1. Heterogeneity in Digital Transformation Types

To further investigate the differential impact of digital transformation on environmental performance, we use sub-dimensional digital transformation variables in our regression analysis. Table 10 presents the regression results for the core independent variables based on the four sub-dimensions: digital technology, business digital transformation, manufacturing digital transformation, and operational digital transformation.

Columns (1) to (4) indicate that all four sub-dimensions of digital transformation significantly enhance environmental performance at the 1% level, though the economic impacts vary. *Ceteris paribus*, a one-standard deviation increase in manufacturing digital transformation results in a 1.35% improvement in environmental performance, followed by technological digital transformation at 0.92%, operational digital transformation at 0.83%, and business digital transformation at 0.66%. The findings suggest that the benefits derived from different directions of digital transformation are not uniform. Further, a relative importance comparison of the four dimensions is provided in column (4). In this regression, business digital transformation loses significance, while the dimensions with significant positive effects, ranked by impact strength, are

manufacturing digital transformation (1.08%), followed by operational digital transformation (0.52%) and technological digital transformation (0.48%). Thus, compared to business digital transformation, digital transformations related to technology, processes, and production serve as more pivotal levers for enhancing environmental performance.

**Table 10: Impact Across Digital Transformation Sub-dimensions**

|                                           | <i>EScore<sub>i,t</sub></i> |                      |                      |                      |                      |
|-------------------------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------|
|                                           | (1)                         | (2)                  | (3)                  | (4)                  | (5)                  |
| <i>LnDT_Technology<sub>i,t-1</sub></i>    | 0.448***<br>(0.062)         |                      |                      |                      | 0.235***<br>(0.064)  |
| <i>LnDT_Business<sub>i,t-1</sub></i>      |                             | 0.294***<br>(0.056)  |                      |                      | 0.072<br>(0.059)     |
| <i>LnDT_Manufacturing<sub>i,t-1</sub></i> |                             |                      | 0.671***<br>(0.058)  |                      | 0.535***<br>(0.061)  |
| <i>LnDT_Operation<sub>i,t-1</sub></i>     |                             |                      |                      | 0.488***<br>(0.057)  | 0.305***<br>(0.058)  |
| Constant                                  | 51.989***<br>(1.223)        | 51.681***<br>(1.225) | 51.505***<br>(1.218) | 51.203***<br>(1.232) | 51.828***<br>(1.216) |
| Firm fixed effects                        | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effects                        | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations                              | 38,249                      | 38,249               | 38,249               | 38,249               | 38,249               |
| R <sup>2</sup>                            | 0.634                       | 0.633                | 0.635                | 0.634                | 0.637                |

Notes: This table analyzes the impact of different sub-dimensions of corporate digital transformation on environmental performance and compares their relative importance. Columns (1)–(4) present results for technological digital transformation (*LnDT\_Technology*), business digital transformation (*LnDT\_Business*), manufacturing digital transformation (*LnDT\_Manufacturing*), and operational digital transformation (*LnDT\_Operation*), respectively. Column (5) includes all four dimensions simultaneously. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors, reported in parentheses, are clustered by firm. \*\*\*, \*\*, and \* indicate statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

## 2. Heterogeneity in Firm Characteristics

To assess the significance of the effect of corporate digital transformation on environmental performance across diverse firms, we further subdivide the sample into two groups based on three firm characteristics. Specifically, firms are categorized by size using the median of *LnMktCap*, resulting in a large group and a small group. In terms of ownership type, we differentiate between state-owned enterprises (SOEs) and non-state-owned enterprises (non-SOEs). For regional economic development, the sample is further classified into high GDP and low GDP groups using the median of city-level GDP.

Table 11 presents the regression results, indicating that corporate digital transformation positively influences environmental performance across all groups, with significance at the 1% level. Addressing concerns regarding the digital divide, we argue that irrespective of firm size, ownership type, or regional economic status, all enterprises can effectively leverage digital transformation to enhance environmental performance.

**Table 11: Impact Across Different Firm Characteristics**

|                             | <i>EScore<sub>i,t</sub></i> |                      |                      |                      |                      |                      |
|-----------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                             | (1)<br>Large                | (2)<br>Small         | (3)<br>SOE           | (4)<br>Non-SOE       | (5)<br>High GDP      | (6)<br>Low GDP       |
| <i>LnDT<sub>i,t-1</sub></i> | 0.646***<br>(0.086)         | 0.624***<br>(0.090)  | 0.865***<br>(0.104)  | 0.515***<br>(0.084)  | 0.614***<br>(0.087)  | 0.675***<br>(0.095)  |
| Constant                    | 43.139***<br>(2.055)        | 53.480***<br>(2.210) | 47.360***<br>(2.406) | 51.752***<br>(1.594) | 55.104***<br>(1.658) | 47.162***<br>(1.815) |
| Control variables           | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Firm fixed effects          | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effects          | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations                | 20,703                      | 16,969               | 13,918               | 21,776               | 20,403               | 17,331               |
| R <sup>2</sup>              | 0.642                       | 0.682                | 0.643                | 0.680                | 0.638                | 0.664                |

GDP = gross domestic product, SOE = state-owned enterprise.

Notes: This table examines whether the effect of corporate digital transformation on environmental performance holds significant across firms with varying characteristics, including size (columns 1 and 2), ownership type (columns 3 and 4), and regional economic development (columns 5 and 6). Size is categorized into large and small based on the median of *LnMktCap*. Ownership type differentiates between SOEs and non-SOEs, while regional economic development is divided into high GDP and low GDP based on the median city GDP. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors, in parentheses, are clustered by firm. \*\*\*, \*\*, and \* denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

Further, we analyze the heterogeneity among the three firm characteristics in conjunction with the four sub-dimensions of digital transformation to examine whether different types of firms exhibit similar choices in their digital transformation directions. The regression results are reported in Appendix 3, with Panel A focusing on size-based analysis, Panel B distinguishing between SOEs and non-SOEs, and Panel C based on regional economic development. The results reveal that, for large enterprises, SOEs, and firms located in high GDP regions, the ranking of sub-dimensions of digital transformation is consistent, with manufacturing digital transformation dominating the strongest impact, significantly outweighing other dimensions, followed by operational digital transformation, technological digital transformation, and business digital transformation. Notably, in small enterprises, operational digital transformation exhibits the strongest effect, while in non-SOEs and firms in low GDP areas, operational digital transformation is overshadowed by technological digital transformation. Consequently, managers and local governments must tailor their digital transformation strategies or policies according to firm characteristics and external environmental differences.

### 3. Heterogeneity in Sectors

To further examine the differential effects of corporate digital transformation on environmental performance across sectors, we conduct sector-specific standardized regressions and rank the results for comparison. Table 12 presents regression results for each sector, ranked by the significance and magnitude of the *LnDT* coefficient from highest to lowest.

In terms of digital transformation-environmental performance synergy, sectors such as Electricity, Heat, Gas, and Water Production and Supply, Transportation, Storage, and Postal Services, Construction, Real Estate, and Manufacturing appear to benefit more from digital transformation initiatives in enhancing environmental performance. These industries, often characterized by high environmental impacts and capital-intensive operations, have stronger incentives and capacities to leverage digital technologies for sustainable practices. This may be because of more stringent



environmental regulations and higher stakeholder scrutiny. As shown in Figure 3, Panel B, these industries also tend to focus on manufacturing digital transformation, which aligns closely with their operational needs.

In contrast, sectors like *Financial Services, Wholesale and Retail, Accommodation and Catering, and Agriculture, Forestry, Animal Husbandry, and Fishery* exhibit either insignificant or negative coefficients, suggesting limited or potentially adverse effects of digital transformation on environmental performance. Notably, while these sectors often show high levels of business and operational digital transformation, their digital transformation initiatives may lack direct efficiency gains in resource and energy use, limiting their impact on environmental performance.

These findings underscore sector-specific heterogeneity, suggesting that the environmental benefits of digital transformation are not uniform across industries. For sectors with lower environmental intensity, digital initiatives may not translate as effectively into environmental performance improvements, highlighting the need for tailored digital transformation strategies to optimize sector-specific environmental outcomes.

**Table 12: Impact Across Different Sectors**

| Sector Name                                                    | Beta     | <i>LnDT_std<sub>i,t-1</sub></i> |        | R <sup>2</sup> |
|----------------------------------------------------------------|----------|---------------------------------|--------|----------------|
|                                                                |          | Std. Err.                       | Obs.   |                |
| Electricity, heat, gas, and water production and supply        | 0.232*** | -0.035                          | 1,364  | 0.733          |
| Transportation, storage, and postal services                   | 0.170*** | -0.047                          | 1,136  | 0.625          |
| Construction                                                   | 0.154*** | -0.044                          | 970    | 0.657          |
| Real estate                                                    | 0.143*** | -0.037                          | 1,464  | 0.634          |
| Manufacturing                                                  | 0.110*** | -0.014                          | 23,315 | 0.636          |
| Information transmission, software, and IT Services            | 0.130**  | -0.055                          | 2,834  | 0.591          |
| Culture, sports, and entertainment                             | 0.209*   | -0.105                          | 546    | 0.552          |
| Mining                                                         | 0.108*   | -0.059                          | 905    | 0.702          |
| Health and social work                                         | 0.307    | -0.222                          | 145    | 0.619          |
| Education                                                      | 0.107    | -0.092                          | 115    | 0.799          |
| Financial services                                             | 0.070    | -0.073                          | 1,132  | 0.624          |
| Leasing and business services                                  | 0.034    | -0.099                          | 603    | 0.579          |
| Agriculture, forestry, animal husbandry, and fishery           | 0.020    | -0.067                          | 498    | 0.604          |
| Water conservancy, environment, and public facility management | 0.004    | -0.074                          | 632    | 0.731          |
| Comprehensive                                                  | -0.002   | -0.123                          | 173    | 0.568          |
| Wholesale and r                                                | -0.022   | -0.062                          | 1,939  | 0.589          |
| Accommodation and catering                                     | -0.088   | -0.153                          | 83     | 0.775          |
| Scientific research and technical services                     | -0.115   | -0.083                          | 391    | 0.685          |

Obs. = observations, Std. Err = standard error.

Notes: This table examines the differential effects of corporate digital transformation on environmental performance across various sectors. The classification of sectors is based on the 2012 CSRC Industry Classification, which includes 19 categories. The "Residential, Repair, and Other Services" sector is excluded because of insufficient observations. To enable direct comparison of the regression coefficients, continuous key variables are standardized separately within each sector. The table presents the regression results for each sector, ranked by the significance and magnitude of the *LnDT* coefficient (Beta), from highest to lowest. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors are clustered by firm. \*\*\*, \*\*, and \* denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

## V. CONCLUSION

This paper examines the impact of corporate digital transformation on environmental performance. Using textual analysis of firms' annual reports, we first construct a four-dimensional measure of corporate digital transformation, encompassing technological, business, manufacturing, and operational aspects. Analyzing a sample of Chinese A-share listed firms from 2009 to 2022, we demonstrate that corporate digital transformation significantly enhances environmental performance. This causal relationship remains robust under endogeneity tests and robustness tests. Mechanism tests reveal that green innovation, strengthened environmental monitoring and communication, increased environmental information transparency, and greater external attention are the underlying channels. Notably, environmental information transparency exerts the strongest influence in this context. Moreover, heterogeneity analyses compare the influence of the four digital transformation sub-dimensions, with manufacturing digital transformation having the most significant impact, followed by operational and technological digital transformation. The research also categorizes firms by size, ownership type, and regional economic development, finding that the positive effect of digital transformation on environmental performance holds across all groups. It is worth noting that economic benefits vary across different types of firms based on the specific digital transformation sub-dimension adopted.

Based on the findings of this paper, the following policy recommendations are proposed. First, the integration of digitalization into environmental sustainability initiatives should be actively promoted and prioritized in both policy-making and corporate strategy development. Specifically, as the government advances digital economic growth and builds a nationwide high-level digital infrastructure, it should also incorporate digital transformation into national and local environmental policies. For instance, in August 2024, the State Council of the PRC issued the *Guidelines for Coordinated Digital and Green Transformation*, which highlight the strategic importance of synchronized digital and green development and provide sector-specific implementation strategies and best practices. Therefore, raising awareness and offering practical, actionable guidance is critical to help companies align their financial and environmental objectives. Digital technologies are not merely technical tools; they serve as strategic enablers. For example, the application of technologies such as the IoT, big data, and blockchain can enhance pollution monitoring, carbon footprint tracking, and resource efficiency, thereby improving corporate environmental compliance and competitiveness. With appropriate policy support, firms can progressively develop a comprehensive understanding of digital transformation as a key driver of both financial performance and environmental sustainability.

Second, recognizing that industries and regions exhibit diverse needs and capabilities in the context of digital green synergy, policymakers should establish targeted guidelines that address these variations. Specifically, policymakers should consider the characteristics of the region, industry, and individual firms to classify and guide different types of enterprises in adopting distinct digital transformation strategies and approaches, as well as in leveraging digital technologies for green transformation through various pathways. By developing sector-specific and characteristic relevant guidance, policymakers can facilitate more effective digital transformation efforts, ensuring that all firms, regardless of their current digital maturity, can utilize digital technologies to achieve their environmental objectives. This tailored approach not only enhances policy effectiveness, but also promotes equitable access to resources and support across diverse business environments.

Third, strengthening attention and collaboration is essential for fostering leadership in digital and green transformation. The government should encourage companies to enhance their competitiveness through green innovation and digital solutions while cultivating industry leaders in digital green initiatives. These firms should receive incentives and recognition as models to

inspire broader participation in this process. Additionally, cross-industry and cross-sector collaboration must be actively supported. For instance, partnerships between environmental technology companies and digital firms should be encouraged to jointly develop smart environmental solutions. Further, collaboration between multinational and local enterprises can facilitate the sharing of global best practices, thereby advancing the agenda for sustainable development on a worldwide scale.

While this study quantitatively examines the relationship between corporate digital transformation and environmental performance in the PRC, it is important to acknowledge certain limitations that may guide future research endeavors. This investigation primarily focuses on the direct effects and transmission mechanisms of digital transformation on environmental performance. However, it does not explore the potential for nonlinear relationships between these variables. Therefore, further studies could consider developing theoretical models to elucidate the mechanisms and pathways that may characterize such nonlinear dynamics. Ultimately, this paper contributes new insights into the critical role of digitalization in promoting sustainability for firms.

## Appendix 1: Definitions of Variables

| Variable                  | Definition                                                                                                                                            |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>EScore</i>             | Environmental scores from the Sino-Securities Index ESG ratings, ranging from 0 to 100                                                                |
| <i>EScore_wind</i>        | Environmental scores from the Wind ESG Ratings, ranging from 0 to 10                                                                                  |
| <i>EnvTax</i>             | Quartile ranking of a firm's environmental protection tax (scaled by total assets) within the same industry and year, with values ranging from 1 to 4 |
| <i>LnDT</i>               | Natural logarithm of one plus the frequency of terms related to digital transformation in firms' annual reports                                       |
| <i>LnDT_Technology</i>    | Natural logarithm of one plus the frequency of terms related to digital technologies in firms' annual reports                                         |
| <i>LnDT_Business</i>      | Natural logarithm of one plus the frequency of terms related to business digital transformation in firms' annual reports                              |
| <i>LnDT_Manufacturing</i> | Natural logarithm of one plus the frequency of terms related to manufacturing digital transformation in firms' annual reports                         |
| <i>LnDT_Operation</i>     | Natural logarithm of one plus the frequency of terms related to operational digital transformation in firms' annual reports                           |
| <i>GPatent</i>            | Number of green patents filed (and eventually granted) by the enterprise, its subsidiaries, and joint ventures                                        |
| <i>EnvResponse</i>        | Establishment of an emergency response mechanism for major environmental events, coded as 1 if established and 0 otherwise                            |
| <i>EnvDisclosure</i>      | Disclosure of relevant environmental information in public reports, coded as 1 if disclosed and 0 otherwise                                           |
| <i>AnaCoverage</i>        | Number of analysts (or teams) conducting tracking analysis on the company during the fiscal year                                                      |
| <i>LnMktCap</i>           | Natural logarithm of the annual circulating market capitalization                                                                                     |
| <i>ROE</i>                | Net profit divided by average balance of shareholders' equity                                                                                         |
| <i>Leverage</i>           | Total liabilities divided by total assets                                                                                                             |
| <i>LnListAge</i>          | Natural logarithm of one plus the difference between the fiscal year and the listing year of the firm                                                 |

ESG = environmental, social, and governance; ROE = return on equity.

Source: Authors' compilation

## Appendix 2: Correlation Matrix of Key Variables

|                           | <i>EScore</i> | <i>LnDT</i> | <i>LnDT_Technology</i> | <i>LnDT_Business</i> | <i>LnDT_Manufacturing</i> | <i>LnDT_Operation</i> | <i>GPatent</i> | <i>EnvResponse</i> | <i>EnvDisclosure</i> | <i>AnaCoverage</i> |
|---------------------------|---------------|-------------|------------------------|----------------------|---------------------------|-----------------------|----------------|--------------------|----------------------|--------------------|
| <i>EScore</i>             |               | 0.075       | 0.038                  | 0.050                | 0.102                     | 0.082                 | 0.161          | 0.160              | 0.202                | 0.091              |
| <i>LnDT</i>               | 0.148         |             | 0.734                  | 0.791                | 0.766                     | 0.715                 | 0.104          | -0.049             | 0.045                | 0.063              |
| <i>LnDT_Technology</i>    | 0.106         | 0.757       |                        | 0.625                | 0.507                     | 0.479                 | 0.067          | -0.047             | 0.039                | 0.042              |
| <i>LnDT_Business</i>      | 0.088         | 0.829       | 0.666                  |                      | 0.436                     | 0.459                 | 0.050          | -0.094             | 0.029                | 0.070              |
| <i>LnDT_Manufacturing</i> | 0.199         | 0.748       | 0.542                  | 0.440                |                           | 0.431                 | 0.190          | 0.031              | 0.081                | 0.016              |
| <i>LnDT_Operation</i>     | 0.142         | 0.748       | 0.569                  | 0.505                | 0.497                     |                       | 0.046          | -0.051             | 0.021                | 0.067              |
| <i>GPatent</i>            | 0.124         | 0.066       | 0.032                  | 0.054                | 0.105                     | 0.026                 |                | 0.070              | 0.100                | 0.167              |
| <i>EnvResponse</i>        | 0.190         | 0.038       | -0.028                 | -0.033               | 0.143                     | 0.029                 | 0.083          |                    | 0.185                | 0.048              |
| <i>EnvDisclosure</i>      | 0.230         | 0.166       | 0.113                  | 0.093                | 0.202                     | 0.120                 | 0.076          | 0.205              |                      | 0.048              |
| <i>AnaCoverage</i>        | 0.111         | 0.101       | 0.054                  | 0.104                | 0.033                     | 0.094                 | 0.220          | 0.060              | 0.075                |                    |

Notes: This table reports the correlations between the key variables, with Pearson (Spearman) correlation coefficients above (below) the diagonal. Detailed definitions of variables are in the Appendix 1.

Source: Authors' compilation.

### Appendix 3: Sub-Dimensional Digital Transformation Impact Across Different Firm Characteristics

| Panel A: Subgroups Based on Firm Size     |                             |                      |                      |                      |                      |                      |                      |                      |
|-------------------------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                           | <i>EScore<sub>i,t</sub></i> |                      |                      |                      |                      |                      |                      |                      |
|                                           | (1)<br>Large                | (2)<br>Small         | (3)<br>Large         | (4)<br>Small         | (5)<br>Large         | (6)<br>Small         | (7)<br>Large         | (8)<br>Small         |
| <i>LnDT_Technology<sub>i,t-1</sub></i>    | 0.355***<br>(0.082)         | 0.521***<br>(0.101)  |                      |                      |                      |                      |                      |                      |
| <i>LnDT_Business<sub>i,t-1</sub></i>      |                             |                      | 0.276***<br>(0.071)  | 0.315***<br>(0.082)  |                      |                      |                      |                      |
| <i>LnDT_Manufacturing<sub>i,t-1</sub></i> |                             |                      |                      |                      | 0.678***<br>(0.079)  | 0.567***<br>(0.083)  |                      |                      |
| <i>LnDT_Operation<sub>i,t-1</sub></i>     |                             |                      |                      |                      |                      |                      | 0.461***<br>(0.079)  | 0.604***<br>(0.082)  |
| Constant                                  | 43.973***<br>(2.060)        | 54.085***<br>(2.204) | 43.774***<br>(2.058) | 53.935***<br>(2.217) | 43.377***<br>(2.047) | 53.924***<br>(2.204) | 43.209***<br>(2.060) | 53.415***<br>(2.206) |
| Control variables                         | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Firm fixed effects                        | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effects                        | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations                              | 20,703                      | 16,969               | 20,703               | 16,969               | 20,703               | 16,969               | 20,703               | 16,969               |
| R <sup>2</sup>                            | 0.641                       | 0.681                | 0.640                | 0.680                | 0.643                | 0.681                | 0.641                | 0.682                |

| Panel B: Subgroups Based on Firm Ownership Types |                             |                      |                      |                      |                      |                      |                      |                      |
|--------------------------------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                                  | <i>EScore<sub>i,t</sub></i> |                      |                      |                      |                      |                      |                      |                      |
|                                                  | (1)<br>SOE                  | (2)<br>Non-SOE       | (3)<br>SOE           | (4)<br>Non-SOE       | (5)<br>SOE           | (6)<br>Non-SOE       | (7)<br>SOE           | (8)<br>Non-SOE       |
| <i>LnDT_Technology<sub>i,t-1</sub></i>           | 0.497***<br>(0.111)         | 0.444***<br>(0.083)  |                      |                      |                      |                      |                      |                      |
| <i>LnDT_Business<sub>i,t-1</sub></i>             |                             |                      | 0.477***<br>(0.094)  | 0.213***<br>(0.073)  |                      |                      |                      |                      |
| <i>LnDT_Manufacturing<sub>i,t-1</sub></i>        |                             |                      |                      |                      | 0.775***<br>(0.096)  | 0.682***<br>(0.077)  |                      |                      |
| <i>LnDT_Operation<sub>i,t-1</sub></i>            |                             |                      |                      |                      |                      |                      | 0.625***<br>(0.094)  | 0.440***<br>(0.078)  |
| Constant                                         | 48.436***<br>(2.411)        | 52.477***<br>(1.582) | 48.408***<br>(2.412) | 52.129***<br>(1.591) | 47.847***<br>(2.410) | 52.094***<br>(1.571) | 47.602***<br>(2.429) | 51.796***<br>(1.593) |
| Control variables                                | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Firm fixed effects                               | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effects                               | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations                                     | 13,918                      | 21,776               | 13,918               | 21,776               | 13,918               | 21,776               | 13,918               | 21,776               |
| R <sup>2</sup>                                   | 0.640                       | 0.680                | 0.640                | 0.679                | 0.642                | 0.681                | 0.641                | 0.680                |

Continued on the next page

**Panel C: Subgroups Based on Regional Economic Development**

| Variables                                 | <i>EScore<sub>i,t</sub></i> |                      |                      |                      |                      |                      |                      |                      |
|-------------------------------------------|-----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                           | (1)<br>High GDP             | (2)<br>Low GDP       | (3)<br>High GDP      | (4)<br>Low GDP       | (5)<br>High GDP      | (6)<br>Low GDP       | (7)<br>High GDP      | (8)<br>Low GDP       |
| <i>LnDT_Technology<sub>i,t-1</sub></i>    | 0.410***<br>(0.085)         | 0.509***<br>(0.094)  |                      |                      |                      |                      |                      |                      |
| <i>LnDT_Business<sub>i,t-1</sub></i>      |                             |                      | 0.280***<br>(0.078)  | 0.334***<br>(0.084)  |                      |                      |                      |                      |
| <i>LnDT_Manufacturing<sub>i,t-1</sub></i> |                             |                      |                      |                      | 0.637***<br>(0.080)  | 0.739***<br>(0.087)  |                      |                      |
| <i>LnDT_Operation<sub>i,t-1</sub></i>     |                             |                      |                      |                      |                      |                      | 0.555***<br>(0.080)  | 0.473***<br>(0.083)  |
| Constant                                  | 55.671***<br>(1.651)        | 48.097***<br>(1.813) | 55.403***<br>(1.651) | 47.732***<br>(1.813) | 55.350***<br>(1.644) | 47.364***<br>(1.802) | 55.000***<br>(1.667) | 47.117***<br>(1.814) |
| Control variables                         | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Firm fixed effects                        | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effects                        | Yes                         | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations                              | 20,403                      | 17,331               | 20,403               | 17,331               | 20,403               | 17,331               | 20,403               | 17,331               |
| R-squared                                 | 0.637                       | 0.663                | 0.637                | 0.662                | 0.639                | 0.664                | 0.638                | 0.662                |

GDP = gross domestic product, SOE = state-owned enterprise.

Notes: This table examines how the effects of corporate digital transformation sub-dimensions on environmental performance differ across firms based on size, ownership type, and regional economic development. The four dimensions of digital transformation are technological digital transformation (*LnDT\_Technology*), business digital transformation (*LnDT\_Business*), manufacturing digital transformation (*LnDT\_Manufacturing*), and operational digital transformation (*LnDT\_Operation*). Panel A analyzes size, categorizing firms into large and small based on the median of *LnMktCap*. Panel B differentiates between state-owned enterprises and non-state-owned enterprises. Panel C classifies firms into high GDP and low GDP groups using the median city GDP. Control variables and fixed effects are consistent with the baseline model, as shown in column (2) of Table 2. Robust standard errors, in parentheses, are clustered by firm. \*\*\*, \*\*, and \* denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Source: Authors' calculations.

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# Corporate Digital Transformation and Environmental Performance

*Evidence from the People's Republic of China*

This paper examines how digital transformation influences corporate environmental performance. Using firm-level data from the People's Republic of China between 2009 and 2022, the study finds that firms' digital transformation significantly improves their environmental performance scores. Improved green innovation, enhanced environmental monitoring and transparency, and increased external attention are identified as possible channels underlying this relationship. The results offer policy insights on accelerating digital transformation to support the green transition.

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